Journal of Environmental Chemical Engineering 14 (2026) 123307

Contents lists available at ScienceDirect

Journal of Environmental Chemical Engineering

journal homepage: www.elsevier.com/locate/jece

Enhancing anaerobic co-digestion of poultry litter and wheat straw via integrated microaeration, biochar, and nanoparticle strategies: Kinetic, microbial, and economic insights

Yuanhang Zhan [a,b], Yiting Xiao [a], Sarah Wu [c], Zhiwu Wang [d], Jun Zhu [a,*]

[a] Department of Biological and Agricultural Engineering, University of Arkansas, Fayetteville, AR 72701, USA
[b] Department of Bioproducts and Biosystems Engineering, University of Minnesota, Saint Paul, MN 55108, USA
[c] Department of Chemical and Biological Engineering, University of Idaho, Moscow, ID 83844, USA
[d] Department of Biological Systems Engineering, Virginia Tech, Blacksburg, VA 24060, USA

ARTICLE INFO

Keywords:
Specific methane yield
Digestate quality
Kinetic modeling analysis
Microbial community shifts
Methanogenic functional prediction
Cost-benefit analysis

ABSTRACT

This study systematically evaluated microaeration (M), biochar (B), and nanoparticle (N) amendments and their combinations for enhancing anaerobic co-digestion (Co-AD) of poultry litter with wheat straw. All strategies improved specific methane yield compared to the control (274.4 NmL CH_4/g VS), with the biochar-nanoparticle combination (B+N) achieving the optimal performance (324.8 ± 3.8 NmL CH_4/g VS), representing an 18.4% improvement. B+N achieved superior substrate degradation with 80.2% total solids and 86.3% volatile solids removal, exceeding the control by 38.6% and 24.2%, respectively. Kinetic modeling using the cone model and transfer model accurately simulated the methane production dynamics ($R^2 > 0.9950$), revealing distinct enhancement mechanisms. M most effectively increased the hydrolysis rate by 12.0% and maximum methane production rate by 19.5%, B enhanced ultimate methane potential, while N most effectively shortened the lag phase to 0.72–0.74 days. Microbial community analysis revealed the mechanistic fundamentals: all strategies enriched the acetoclastic methanogen Methanosaeta, with B+N achieving the highest archaeal abundance (approximately 2.0%). M enriched hydrolytic bacteria enhancing substrate degradation, B facilitated direct interspecies electron transfer (DIET) and microbial colonization while enriching syntrophic bacteria (Syntrophomonas, Syntrophobacter), and N supplied essential trace elements and promoted DIET between syntrophic partners. Cost-benefit analysis based on a simulated continuous reactor framework demonstrated that B+N achieved the optimal long-term economics with break-even at 90–120 days, making it most suitable for sustained operations. This study establishes an integrated kinetic-microbiological-economic framework for Co-AD process optimization, potentially providing practical guidance for industrial-scale implementation and advancing the circular bioeconomy through improved agricultural waste valorization.

1. Introduction

Anaerobic digestion (AD) of agricultural wastes has emerged as a sustainable technology for simultaneously addressing environmental pollution and generating renewable energy through methane-rich biogas production [1,2]. The global imperative to transition toward circular bioeconomy models has intensified interest in AD systems that convert waste biomass into valuable energy while reducing greenhouse gas emissions from waste disposal. Anaerobic co-digestion (Co-AD) of animal manures with carbon-rich agricultural residues offers multiple advantages, including improved methane yield, enhanced process stability, better nutrient retention in digestate for agricultural applications, and optimized feedstock utilization compared to mono-digestion [3–5]. However, the recalcitrance of lignocellulosic substrates and potential inhibitory compounds in animal wastes (ammonia, volatile fatty acids (VFAs), and hydrogen sulfide) often limit hydrolysis efficiency and methanogenic activity, adversely affecting process performance and economic viability. Advanced enhancement strategies targeting these metabolic bottlenecks have been developed to overcome these limitations to promote Co-AD systems.

Microaeration (abbreviated as M), involving controlled dosing of small quantities of air or oxygen (typically <1% of biogas production)

* Corresponding author.
E-mail address: junzhu@uark.edu (J. Zhu).

https://doi.org/10.1016/j.jece.2026.123307
Received 31 March 2026; Received in revised form 11 May 2026; Accepted 24 May 2026
Available online 26 May 2026
2213-3437/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

into anaerobic reactors [6], has demonstrated potential for enhancing hydrolysis rates through selective enrichment of facultative anaerobic bacteria, improving process stability at high organic loading rates by preventing VFA accumulation, and mitigating hydrogen sulfide toxicity through partial oxidation to elemental sulfur or sulfate [7–9]. The introduction of facultative microorganisms under microaerobic conditions can accelerate the degradation of complex organic polymers, particularly recalcitrant lignocellulosic materials, through enhanced extracellular enzyme production while maintaining overall anaerobic conditions favorable for methanogenesis in deeper reactor zones.

Biochar (abbreviated as B), a carbon-rich porous material produced through pyrolysis of biomass under oxygen-limited conditions (typically 300–700°C), has been increasingly utilized as an additive in Co-AD systems due to its high specific surface area, hierarchical pore structure, chemical stability, pH-buffering capacity, and unique electrochemical properties [10–13]. B enhances process performance through multiple synergistic mechanisms including providing three-dimensional colonization surfaces for microbial consortia that facilitate spatial organization and protection from environmental stresses, acting as a conductive matrix to facilitate direct interspecies electron transfer (DIET) between syntrophic bacteria and methanogenic archaea, adsorbing and sequestering inhibitory compounds such as ammonia and VFA to maintain favorable process conditions, promoting microbial aggregation and biofilm formation that enhance metabolic cooperation, improving nutrient retention through cation exchange capacity, and buffering pH fluctuations during acidogenic phases [14,15].

Nanoparticles (abbreviated as N), nano-sized materials (1–100 nm) with unique physicochemical properties arising from their high surface-area-to-volume ratios and quantum effects, represent an emerging strategy for Co-AD enhancement with applications spanning trace element supplementation, electron transfer mediation, and catalytic activity stimulation [16]. Metal-based N, particularly iron oxides (Fe_3O_4, Fe_2O_3), nickel, cobalt, and their composites, can serve as readily bioavailable sources of trace elements essential for methanogenic enzyme cofactors such as iron-sulfur clusters in hydrogenases, nickel in methyl-coenzyme M reductase, and cobalt in corrinoid-dependent methyltransferases [17,18]. Beyond nutritional supplementation, N can enhance DIET through increased system conductivity, stimulate enzyme catalytic activity through surface interactions and enhanced substrate accessibility, mitigate inhibitory effects by adsorbing toxic compounds or altering redox conditions, and influence microbial community structure through selective pressures favoring metal-resistant and metal-utilizing microorganisms [19,20].

Although M, B, and N supplementation have individually demonstrated effectiveness in enhancing AD or Co-AD performance, systematic comparative studies evaluating their relative efficiencies under identical operational conditions remain limited. Furthermore, potential synergistic effects from combining these strategies, which may target complementary metabolic bottlenecks or create favorable microenvironments through multiple simultaneous mechanisms, have not been comprehensively investigated. Understanding the complementary enhancement mechanisms is critical for developing optimized, cost-effective strategies for industrial-scale implementation that balance performance improvements against operational costs and practical constraints. Additionally, the microbial community responses to these enhancement strategies, including shifts in taxonomic composition, functional gene expression, and syntrophic network structure, remain incompletely characterized, limiting mechanistic understanding and rational design of enhancement protocols.

This study systematically compared M, B, and ferric oxide (Fe_3O_4) nanoparticles (N) as an effective nanoparticle that could enhance the AD process [21,22], both individually and in their combinations, for enhancing batch Co-AD of poultry litter (PL) with wheat straw (WS). Methane production performance was assessed through specific methane yield (SMY) and kinetic parameter determination using mathematical models to capture distinct process phases. Well-established empirical kinetic models were applied to all treatment groups using a consistent framework, enabling direct comparison of key parameters (e.g., methane potential, rate, and lag phase) across different enhancement strategies (M, B, N, and their combinations). Substrate degradation efficiency was evaluated through total solids (TS) and volatile solids (TVS) removal, providing insights into organic matter conversion. Microbial community dynamics were comprehensively characterized using high-throughput 16S rRNA gene sequencing to elucidate taxonomic composition, with focused analysis of archaeal populations and predicted methanogenic pathways. Cost-benefit analysis based on a simulated continuous anaerobic sequencing batch reactor (ASBR) framework evaluated economic viability. Unlike previous studies that typically examine these approaches individually or in limited combinations, this work evaluated single, binary, and ternary configurations under consistent conditions, enabling direct comparison of their interaction effects. This approach provided a more comprehensive understanding of how these strategies independently and collectively influenced different stages of the Co-AD process, as reflected by process performance, kinetic behavior, and microbial community dynamics. The findings offer practical guidance for advancing Co-AD technology toward sustainable waste valorization and renewable energy production, contributing to circular bioeconomy development and climate change mitigation through enhanced bioenergy recovery from agricultural waste.

2. Materials and methods

2.1. Substrates, inoculum, biochar, and Fe_3O_4 nanoparticles

Poultry litter (PL) was collected from the poultry house beddings on a poultry farm (University of Arkansas - Applied Broiler Research Farm, AR, US). The raw wheat straw (WS) was collected from a university grass farm (Agricultural Research & Extension Center, AR, US). The raw substrates were mechanically cut separately into small pieces, using two identical grinders (Food Processor, Hamilton Beach, Inc., VA, US). The results of the physicochemical properties of the pretreated PL and WS are shown in Table 1. As shown, PL had a relatively lower carbon-to-nitrogen ratio (C/N, 8) than WS (50). Thus, the mixing of PL and WS could formulate a co-substrate with a designed balanced C/N ratio with a value between 8 and 50.

Biochar was manufactured by pyrolysis of wood from recycled materials by Wakefield Biochar (Columbia, USA). Fe_3O_4 nanoparticles used in this study were manufactured by SkySpring Nanomaterials, Inc (Houston, USA). The basic properties of the biochar and the Fe_3O_4 nanoparticles are summarized in Table 1. A scanning electron microscope (SEM) was used to characterize the structure of biochar and Fe_3O_4 nanoparticles. The images at different magnifications, including 500μm, 100μm, and 50μm for biochar and 1μm, 500 nm, and 100 nm for Fe_3O_4 nanoparticles, are depicted in Fig. S1. SEM images revealed that the biochar possesses a well-developed porous, honeycomb-like structure with interconnected channels, providing a high surface area for microbial attachment and material immobilization. Fe_3O_4 nanoparticles exhibit a relatively uniform, near-spherical morphology with consistent particle sizes and form homogeneously distributed aggregates without pronounced size heterogeneity.

The microbial inoculum sludge collected from a laboratory-scale anaerobic digester was first sieved through a 1.40 mm mesh to remove coarse particles. The sieved sludge was subsequently pre-incubated at 35 ± 2 °C or two weeks until biogas production became negligible, indicating exhaustion of readily biodegradable substrates and minimizing background methane interference. After pre-treatment, the sludge was stored in a tightly sealed, opaque container at 0°C and later used as the microbial inoculum for the Co-AD experiments. The main characteristics of the inoculum sludge are presented in Table 1.

Table 1
Physicochemical properties of the substrate and the inoculum sludge.

Properties	Poultry litter	What straw	Biochar	Fe_3O_4 nanoparticles	Inoculum sludge
TS	(87.18 ± 1.30) %	(98.15 ± 0.25) %	-	-	(2.84 ± 0.13) %
TVS	(62.63 ± 1.00) %	(89.90 ± 0.67) %	-	-	(18.49 ± 0.26) g/L
TC (%)	26.04	42.10	86–88%	-	-
pH	8.0	6.3	10.6–11	-	7.91 ± 0.01
Size	< 0.85 mm	< 0.85 mm	1–4 mm	20–30 nm	-
TN	3.12%	0.92%	0.7%	-	(436 ± 6) mg /L
TAN	-	-	-	-	(383 ± 7) mg /L

Note: TS was total solids level (%); TVS was total volatile solids level (% substrate or g/L); TC and TN were the total carbon content (%) and total nitrogen content (%), respectively; TAN was the ammonium nitrogen concentration (mg/L).

2.2. Experimental design and methodology

This study systematically evaluated M, B, and N, both individually and in combinations, for enhancing batch Co-AD of PL with WS. Eight experimental groups were established: control with no enhancement (C), single-strategy treatments (M, B, N), dual-strategy combinations (M+B, M+N, B+N), and triple-strategy integration (M+B+N). The operational conditions for each enhancement strategy were optimized based on previous experiments and established literature recommendations [23–25], as summarized in Table 2. M was implemented by slowly injecting a controlled volume of air into batch reactors daily (1.04 mL per day) for the first 4 consecutive days using a syringe to maintain microaerobic conditions while preserving overall anaerobic metabolism. B and N were added with the substrate at reactor initialization to ensure immediate availability throughout the digestion process. Each treatment group was conducted in duplicate to ensure statistical reliability and obtain mean performance values. All operational parameters of the batch Co-AD process were maintained constant across all groups as detailed in the following section, with the applied enhancement strategy being the sole variable. This controlled experimental design enabled rigorous comparison of treatment effects on process performance, including methane production and yield kinetics, substrate degradation efficiency (TS and VS removal), and digestate quality.

2.3. Batch anaerobic co-digestion operation

Identical glass containers with a total volume of 450 mL were used for the Co-AD process. Besides the 8 experimental groups (each in duplicates), blank groups (duplicates) with no substrate addition were also included. Duplicate reactors were employed due to the large number of treatment groups and practical constraints of batch Co-AD experiments. Preliminary trials confirmed consistent reactor performance under controlled conditions, and the observed small standard deviations indicated good reproducibility. Nevertheless, the use of duplicates might limit statistical robustness, and the results were interpreted with caution. The glass containers were fully sealed except for two gas outlets, one of which was connected to a Tedlar® gas bag for biogas collection (Tedlar Bag, CEL Scientific Corp., Cerritos, CA, USA), and the other had a tube reaching the sludge at the bottom of the reactor for nitrogen gas flushing and microaeration.

To begin, 0.60 g of PL and 1.64 g of WS were placed into each reactor with 100 mL of the evenly mixed inoculum sludge to reach a C/N ratio of 25 for the loaded substrate and a $VS_{inoculum}/VS_{substrate}$ ratio of 1 as suggested by previous research [9]. The total substrate TS and VS were 2.13 g and 1.85 g, respectively. Deionized water was added to obtain a total working volume of 200 mL in each reactor, which led to a substrate TS of 2.13% and VS of 18.48 g/L (100 mL). The digestion duration (40 days) was determined by substrate degradation under the selected TS level (2.13%), which is lower than in our previous study (4–12%) and resulted in faster completion of methane production. This TS was chosen to maintain a balanced C/N ratio and a $VS_{inoculum}/VS_{substrate}$ ratio of 1, ensuring stable and comparable conditions across all treatments. Each reactor was then purged with nitrogen for 5 min to remove oxygen from the liquid. The reactors were all housed in an incubator at 35 ± 2 °C mesophilic conditions and run to ferment methane until no measurable biogas was produced. The Co-AD system was mixed manually by shaking the reactors twice daily, at intervals of 12 h. The gas bag of each reactor for biogas collection was replaced daily. The collected gas was measured in volume and analyzed for methane content to determine methane production.

Cumulative methane yield (CMY, NmL CH_4/VS_{added}) in each reactor was calculated through successive summation of daily methane yield normalized to the initial VS loading from the initial day to the corresponding operational day. Specific methane yield (SMY, NmL CH_4/g VS_{added}) was determined as the final CMY upon cessation of significant methane generation. Upon completion of the batch Co-AD process, digestate samples were collected from each reactor for physicochemical characterization to evaluate the performance of the tested strategies. In parallel, sludge samples were also obtained for microbial community analysis.

2.4. Sample analysis

2.4.1. Physicochemical parameters

Total carbon (TC) and total nitrogen (TN) were measured by combustion using Elementar varioMax CN Cube (Elementar, Germany). The mineral elements, including potassium (K), calcium (Ca), and magne-

Table 2
Experimental designed groups for the Co-AD process applied with different strategies (microaeration (M), biochar addition (B), Fe_3O_4 nanoparticles supplementation (N)).

Experimental Group			Strategy parameters					
No.	Name	Strategy	M (mL air/g TS)	M (mL air/ reactor)	B (mg/g TS)	B (mg/ reactor	N (mg/g TS)	N (mg/ reactor)
1	C	Control	0	0	0	0	0	0
2	M	Micro-aeration	2	4.16	0	0	0	0
3	B	Biochar	0	0	100	213	0	0
4	N	Nanoparticles	0	0	0	0	1.5	3.20
5	M+B	Microaeration and biochar	2	4.16	100	213	0	0
6	B+N	Biochar and nanoparticles	0	0	100	23	1.5	3.20
7	M+N	Micro-aeration and nanoparticles	2	4.16	0	0	1.5	3.20
8	M+B+N	Micro-aeration, biochar, and nanoparticles	2	4.16	100	213	1.5	3.20

sium (Mg), were measured using HNO_3 Digest, Spectro ARCOS I ICP (SPECTRO Analytical Instruments, Germany). Moisture content, total solids (TS), and total volatile solids (TVS) were analyzed by thermogravimetric analysis following the standard methods [26], using an oven (BINDER Inc., Bohemia, NY, USA) set at 105℃, and a muffle furnace (Cole-Parmer, Vernon Hills, IL, USA) set at 550℃, respectively. The substrate TS and TVS removal in each treatment group were determined by subtracting the corresponding values of the blank control group from the measured digestate TS and TVS content, as shown in Eq. (1).

$$\text{TS / TVS removal}(\%) = \frac{S_{substrate} * V_{substrate} - (S_{treatment} - S_{blank}) * V_{working}}{S_{substrate} * V_{substrate}} * 100\% \quad (1)$$

where $S_{substrate}$ = concentration of substrate TS or TVS (g/L); $V_{substrate}$ = volume of substrate solution (0.1 L in this study); $S_{treatment}$ = concentration of TS or TVS (g/L) in the final digestate of the treatment group; S_{blank} = concentration of TS or TVS (g/L) in the final digestate of the blank group; $V_{working}$ = working volume of the reactor (0.2 L in this study).

Soluble chemical oxygen demand (sCOD), volatile fatty acids (VFAs), total ammonia nitrogen (TAN), and total alkalinity (TA) of the digestate samples were analyzed using a DR 3900 Hach Spectrophotometer and Hach vials (Hach Company, IA, US) following the standard methods [26]. All the parameters were measured in triplicate, and the mean values with standard deviations were obtained.

Free ammonia nitrogen (FAN) concentration could be calculated according to the equilibrium equation (Eq. (2), [27]).

$$\frac{FAN}{TAN} = \left(1 + \frac{10^{-pH}}{10^{-\left(0.9018 + \frac{2729.92}{T(K)}\right)}}\right)^{-1} \quad (2)$$

where FAN is the concentration (mg/L) of free ammonia nitrogen; TAN is the concentration (mg/L) of total ammonia nitrogen, and T is the temperature (K) of the digestate.

2.4.2. Biogas analysis

The produced biogas in the daily replaced bags was sampled for composition analysis using a Shimadzu Gas Chromatograph (GC 2014, Shimadzu Scientific Instruments, Inc., Maryland, CO, USA) equipped with a thermal conductivity detector (TCD) and a ShinCarbon column (Restek, length 2 m× ID 2 mm). The operation of GC was described by Shen J, et.al (2017) [28]. The volume of the biogas produced daily was measured using a wet gas test meter (Model XMF-1, Shanghai Cixi Instrument Co., Ltd, China), whose value was then normalized before being used for calculation [29].

2.5. Kinetic study

2.5.1. Kinetic models

The methane yield kinetics of the Co-AD process were characterized using four established mathematical models to determine the optimal fit: the first-order kinetic model (FM), the modified Gompertz model (MGM), the cone model (CNM), and the transfer model (TM). FM (Eq. (3)) was first described by Llabrés-Luengo P, et.al (1987) [30], who assumed that hydrolysis was the rate-limiting step that governed the overall process [31,32]. MGM (Eq. (4)) was first used to predict methane production rate in a batch digester corresponding to the specific growth rate of methanogenic bacteria [33], and later became a widely used empirical regression model that described cell density in terms of exponential growth rates and lag phase duration and could be used to identify the maximum methane production potential and lag phase [34]. CNM (Eq. (5)) has been used as an empirical model to estimate the methane yield rate and the maximum cumulative methane yield, showing better fitting performance than other models in methane production from Co-AD [35,36]. TM (Eq. (6)) was also used to fit the methane production data from anaerobic co-digestion processes [37].

The equations for these kinetic models are detailed as follows. The measured CMY(NmL CH_4/g VS_{added}) data in the 8 experimental groups as a function of time were used for simulation. The data for model fitting were the average values of CMY in the duplicate reactors in each group. The kinetic constants of each model were finalized to achieve the minimum sum of squares of residuals.

$$Y(t) = Y_m * \left(1 - e^{-kt}\right) \quad (3)$$

$$Y(t) = Y_m * \exp\left\{-\exp\left[\frac{R_m * e}{Y_m}(\lambda - t) + 1\right]\right\} \quad (4)$$

$$Y(t) = \frac{Y_m}{1 + (kt)^{-n}} \quad (5)$$

$$Y(t) = Y_m * \left\{1 - \exp\left[\frac{-R_m}{Y_m}(t - \lambda)\right]\right\} \quad (6)$$

where Y (t) = cumulative methane yield (NmL CH_4/g VS_{added}) at the AD process operation time t (day); Y_m = specific methane yield of substrates (NmL CH_4/g VS_{added}); e = 2.718; k = hydrolytic rate constant (day^{-1}); t = digestion time (days); R_m = maximum methane production rate (NmL CH_4/g VS_{added}/day); λ = lag phase (days); n = shape constant, which reveals whether there is a lag phase in the reactor.

2.5.2. Kinetic model evaluation

The predictive performance of the kinetic models was evaluated using three statistical metrics: the coefficient of determination (R^2, Eq. (7)), root mean square error (RMSE, Eq. (8)), and Akaike's information criterion (AIC, Eq. (9)). The R^2 value serves as a goodness-of-fit indicator, where values approaching unity indicate superior model predictive capability. The RMSE is a commonly used metric that measures the standard deviation between the predicted and actual values [32]. The model with a lower RMSE value demonstrates a better fit. The calculation of RMSE is presented in the equation below [38]. AIC is based on information technology advancement, where the model with a lower AIC value is considered more likely to be a good fit for the experimental data [39,40]. They can be calculated using the equations below [28].

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = 1 - \frac{\sum_{i=1}^{m}(P_i - M_i)^2}{\sum_{i=1}^{m}(M_i - \overline{M})^2} \quad (7)$$

$$\text{RMSE} = \sqrt{MSE} = \sqrt{\sum_{i=1}^{m}\frac{(P_i - M_i)^2}{m}} \quad (8)$$

$$\text{AIC} = m * \ln\left(\frac{SS_{res}}{m}\right) + 2 * (N + 1) + \frac{2 * (N + 1) * (N + 2)}{(m - N - 2)} \quad (9)$$

where SS_{res} = the sum of squares of residuals; SS_{tot} = the total sum of squares of deviations; P_i = the predicted value of point i by the model; M_i = the measured value of point i; $\overline{M}$ = the mean of the measured value; m = the number of measurements; MSE = mean square error; N = the number of model parameters.

The difference (Dif, %) between the model-predicted value of Y_m (NmL CH_4/g VS $_{added}$) and the measured value of the final CMY (NmL CH_4/g VS $_{added}$) was also used to evaluate the accuracy of the model prediction, which could be calculated using Eq. (10).

$$\text{Dif}(\%) = \frac{Y_m - CMY_o}{CMY_o} * 100\% \quad (10)$$

where Y_m = specific methane yield of substrates (NmL CH_4/g VS $_{added}$) predicted by the model; CMY_o = the observed CMY (NmL CH_4/g VS $_{added}$).

2.6. Microbial community analysis

DNA extractions were performed using the DNeasy PowerLyzer PowerSoil Kit (Qiagen, Germantown, MD). Polymerase chain reaction (PCR) amplification of the fourth and fifth hypervariable (V4-V5) regions of 16S ribosomal RNA (rRNA) was performed. The specific method of performing the microbial community analysis was detailed previously [41]. Qiime (V1.7.0, https://qiime.org/scripts/assign_taxonomy.html) in the Mothur method (v1.39.5) was performed against the SSUrRNA database of SILVA (Release 138.1, https://www.arb-silva.de/)) for species annotation. Software, *R*, was used for statistical analysis of the microbial community structure using the relative abundance of each taxon based on the identification results of the Operational Taxonomic Units (OTUs). The function prediction analysis of the microbial community of the sludge samples was performed according to the 16S rRNA sequencing data using PICRUSt2, which is a bioinformatics software package (https://github.com/picrust/picrust2/wik) for metagenomic function prediction based on the marker gene (16S rRNA in this study). The Metacyc database was referenced, which is a collection of a series of metabolic pathways and enzymes based on the Kyoto Encyclopedia of Genes and Genomes (KEGG) database.

2.7. Cost and benefit analysis based on anaerobic sequencing batch reactor framework

Cost and benefit analysis was conducted by estimating the additional cost needed for the applied enhancing strategy and the gains in energy benefit brought by enhanced methane generation. This study used a framework of ASBR for continuous Co-AD to estimate the cost and benefit balance in a long-term operation based on a previous study [41]. In the previous study, an ASBR with a working volume of 16 L was operated at an OLR of 3.8 g VS/$L_{reactor}$/day under steady-state conditions following successful startup.

In the current study of a batch experiment, micro-aeration was manually conducted, and the cost was difficult to estimate. However, an air pump (Uniclife Air Pump UL40, Uniclife @ Amazon.com) was used for continuous operation [42], and the cost could be calculated based on the electricity consumed since micro-aeration was a continuous process together with ongoing substrate loading. We proposed a working volume of 16 L, with an OLR of 4 g TS/$L_{reactor}$/day. A 12 W air pump and a flow rate of 0.75 LPH (Litter per hour) were used for microaeration, achieving a microaeration rate of 2 mL air /g TS/day. The local electricity price was $0.1067/kWh. The electric cost by microaeration was estimated as $2.4*10^{-4}/day for a continuous operation using an ASBR.

Biochar and nanoparticles were supplemented as a single addition and could potentially be reused or recycled across multiple batch cycles. The cost was estimated as a one-time expense for each additive. The bulk prices for biochar and nanoparticles were $3/kg and $100/kg, respectively. The dosage of biochar and nanoparticles was 1.07 g/$L_{reactor}$ and 0.02 g/$L_{reactor}$ based on reactor working volume, respectively. As a result, the costs of biochar and nanoparticles were $3.20*10^{-3}/$L_{reactor}$ and $3.20*10^{-3}/$L_{reactor}$, respectively. However, the effective lifetime of biochar and nanoparticles depends on their intrinsic properties and the solids retention time (SRT) of the reactor. In this study, the SRT (ranging from 30 to 360 days in increments of 30 days) was used to represent the effective lifetime of the additives and evaluated in the simulations.

Other costs, such as reactor capital investment and operation and maintenance expenses, were not included due to variations in reactor configurations and system complexity. The objective was to directly compare the cost of the added technologies with the associated performance enhancement and to assess their economic viability. The benefit from bioenergy production can be evaluated by calculating the revenue of the observed methane production from the Co-AD experiments. Energy production (E_p) could be estimated by Eq. (11) below [43], and the calculations were on a VS basis.

$$E_p \quad (kWh/g\ VS) = SMY \quad (NmL\ CH_4/g\ TVS) * 34.5 \quad MJ/N\ m^3CH_4 * \frac{1\text{kWh}}{3.6MJ} * \frac{m^3}{10^6 mL} \tag{11}$$

Where E_p = the estimated energy production ($kWh/g\ VS$) based on VS, SMY ($NmLCH_4/g$ TVS $_{added}$) = the final observed cumulative methane yield, TVS (g) = the total volatile solids content in the loaded substrate, TVS (g) = the total volatile solids content in the loaded substrate.

The energy (E_P) can be monetized by applying the conversion efficiency of a CHP (Combined Heat and Power) system. The electricity required for biogas purification was assumed to be 10% of the total electricity via cogeneration, according to Mainardis M, et.al (2019) [44]. A small CHP unit typically operates with an electrical efficiency of about 35%. Thus, the monetary value of the produced energy (V_p) can be estimated by converting methane to electricity based on this efficiency and applying a retail electricity price of US $0.10 kWh^{-1} . The corresponding calculation is expressed in Eq. (12).

$$V_p \quad (US\$/g\ VS) = E_p \quad (kWh/g\ VS) * 35\% * 0.1 \quad US\$/kWh \tag{12}$$

Based on the equations above, the energy value of the assumed continuous ASBR under different operating groups can be estimated using the equation below.

$$\begin{aligned} V_{p-totAL} \quad (US\$) &= V_p \quad (US\$/g\ VS) * VS_{added} \\ &= V_p \quad (US\$/g\ VS) \quad * OLR \quad (g\ VS/L_{reactor}/d) \\ &* \quad Vo_{reactor}(L) \quad * n\ (day) \end{aligned} \tag{13}$$

where V_p is the cash value of the produced energy (E_p ($kWh/g\ VS$) from the substrate VS added ($US\$/g\ VS$), OLR ($g\ VS/L_{reactor}/d$) is the organic loading rate, $Vo_{reactor}$ (L) is the working volume of the reactor, and n is the operational days.

2.8. Statistical analysis

The original data were recorded, and the mean and standard deviation of each dataset were obtained using Microsoft 365 Excel (Microsoft, WA, USA). The kinetic model was fitted using the solver tool in Excel with the target of the minimum value of SS_{res}. The statistical significance was assessed through an analysis of variance (ANOVA) conducted using JMP software (JMP Pro Version 12.0, from SAS Institute Inc. in North Carolina, USA), with a p-value of 0.05. To evaluate performance differences of the Co-AD process applied with different strategies, Tukey's Honest Significant Difference test was employed. Significant differences were indicated and labeled with different characters in alphabetical order, with the highest level starting from "a".

3. Results and discussion

3.1. Performance of the anaerobic co-digestion (Co-AD)

3.1.1. Biomethane generation potential

The profiles of biogas production and the corresponding methane content in different groups are shown in Fig. S2. The temporal evolution of CMY (NmL CH_4/g VS_{added}) and the final SMY (NmL CH_4/g VS_{added}) across different experimental groups are presented in Fig. 1**(a, b)**, respectively. All enhancement strategies achieved significantly higher SMY ($p < 0.05$) compared to the control, confirming that M, B, N, and their combinations enhanced methane yield from the Co-AD process. The B+N treatment achieved the highest SMY (324.8 ± 3.8 NmL CH_4/g VS_{added}), representing an 18.4% improvement over the control. The treatments B, M+B, M+N, and M+B+N demonstrated the second-highest tier of performance, followed by N and M, which exhibited

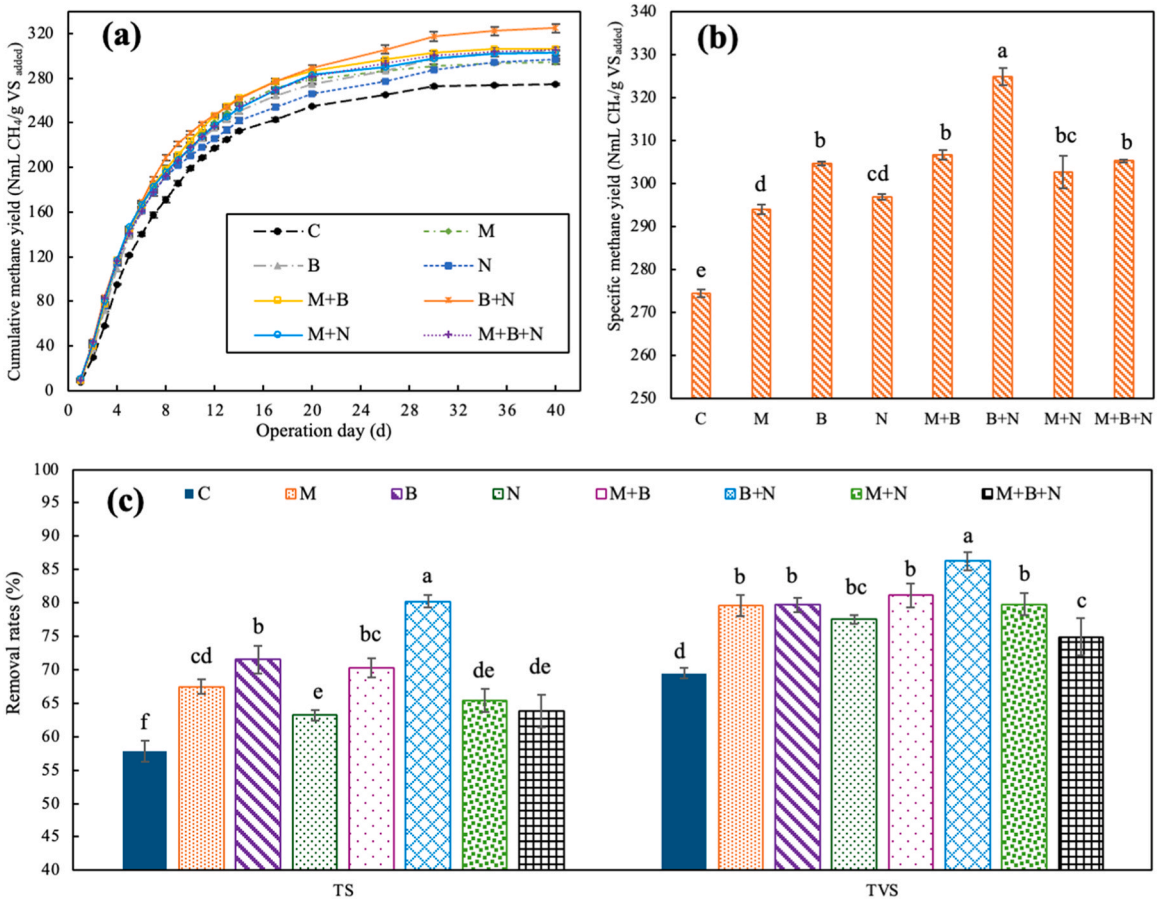


Fig. 1. The comparison of (a) variation of the cumulative methane yield (NmL CH_4/g VS $_{added}$), (b) the specific methane yield (NmL CH_4/g VS $_{added}$), and (c) the removal rates of substrate total solids (TS) and substrate total volatile solids (TVS) in the Co-AD processes receiving different strategies (Data in each with different letters ("a" - highest value and "e" - lowest value) indicate significant differences ($p < 0.05$)).

modest improvements.

Among single-strategy treatments, B yielded the highest SMY (304.7 ± 0.4 NmL CH_4/g VS_{added}, $p < 0.05$), corresponding to an 11.0% enhancement relative to the control. M and N showed comparatively lower improvements of 7.1% and 8.2%, respectively. The superior performance of B can be attributed to its multifunctional role as a microbial colonization substrate, i.e., a conductive matrix facilitating DIET and an adsorbent for inhibitory compounds.

For combined treatments, the B+N strategy achieved significantly higher SMY than either B or N alone, indicating positive synergistic interactions. This synergy likely arises from complementary mechanisms, i.e., biochar provides structural support and conductive surfaces for the growth of microbial consortia, while nanoparticles supply bioavailable trace elements essential for methanogenic enzymes and enhance electron shuttling. The integration of these functions creates an optimized microenvironment that simultaneously addresses multiple metabolic bottlenecks. Conversely, M+B and M+N showed no significant improvement over B and N alone ($p > 0.05$), indicating the absence of synergistic interactions between microaeration and the other treatments. The inability of microaeration to further improve SMY when combined with B or N may could reflect a mechanistic overlap, in which biochar and nanoparticles had already optimized electron transfer and substrate availability, rendering additional oxygen input redundant or even inhibitory. Notably, M+B+N exhibited significantly lower SMY than B+N, demonstrating antagonistic rather than synergistic effects. This antagonism may result from oxygen-induced inhibition of obligate anaerobes, competitive metabolism by facultative aerobes, disrupted electron flow, and/or oxidative passivation of nanoparticles.

3.1.2. *Digestate quality*

The results of the digestate quality in different groups are summarized in Table 3. The final pH values ranged from 7.48 to 7.81 across all experimental groups, falling within the optimal range of 6.5–8.0 for stable AD systems [45], confirming that the process stability was maintained under all treatment conditions. M-containing treatments (M, M+N) exhibited relatively higher pH values, while B-containing treatments (B, B+N) demonstrated lower pH, with intermediate values observed in combined treatments (M+B, M+B+N). This pattern suggests that M tends to elevate pH, possibly through enhanced carbon dioxide stripping or ammonia volatilization, whereas B and N exert pH-buffering or acidifying effects, potentially through surface functional group interactions or enhanced VFA production during intensified substrate degradation.

The concentrations of sCOD, TAN, and VFA in the final digestate reflect the integrated outcomes of substrate hydrolysis, acidogenesis, acetogenesis, and methanogenesis, as well as nitrogen mineralization processes [46,47]. The B+N treatment exhibited relatively elevated levels of sCOD, TAN, VFA, and TA compared to other groups, indicating enhanced organic matter solubilization and mineralization. FAN concentrations remained substantially below inhibitory thresholds of the critical concentration of 1.1 g N/L [48,49] across all treatments, ranging from 3.15 to 5.60 mg N/L. TA, representing the system's acid-neutralizing capacity, is commonly evaluated in conjunction with VFA through the VFA/TA ratio (expressed as equivalent g acetic acid/-equivalent g $CaCO_3$), which serves as a stability indicator for AD systems [50]. The optimal TA between 0.1 and 0.4 indicates favorable operating conditions without risk of acidification or ammonia inhibition [51]. In this study, VFA/TA ratios ranged from 0.17 to 0.20 across all groups,

Table 3
Final digestate quality of the Co-AD process applied with different strategies.

Parameters	C	M	B	N	M+B	B+N	M+N	M+B+N
pH	7.75 ± 0.02b	7.81 ± 0.01a	7.58 ± 0.01e	7.68 ± 0.01c	7.66 ± 0.01c	7.48 ± 0.01 f	7.74 ± 0.02b	7.61 ± 0.02d
sCOD (mg/L)	2093 ± 35d	2098 ± 29d	2103 ± 41d	2323 ± 52ab	2205 ± 26c	2400 ± 22a	2295 ± 52b	2085 ± 13d
TAN (mg/L)	470 ± 3bcd	442 ± 13e	446 ± 6de	479 ± 5b	450 ± 9cde	533 ± 16a	494 ± 9b	475 ± 16bc
FAN (mg/L)	5.20 ± 0.19b	5.60 ± 0.11a	3.29 ± 0.10e	4.49 ± 0.08c	4.06 ± 0.12d	3.15 ± 0.16e	5.34 ± 0.12ab	3.80 ± 0.28d
VFA (mg HAc/L)	337 ± 4c	352 ± 9abc	339 ± 11bc	373 ± 12a	380 ± 22a	354 ± 13abc	374 ± 12a	366 ± 6ab
TA (mg $CaCO_3$/L)	1913 ± 22 cd	1948 ± 59bcd	1928 ± 84 cd	2095 ± 26a	2023 ± 35abc	2055 ± 50ab	1890 ± 59d	1923 ± 25 cd
VFA/TA	0.18 ± 0b	0.18 ± 0.01ab	0.18 ± 0.01b	0.18 ± 0.01b	0.19 ± 0.01ab	0.17 ± 0.01b	0.20 ± 0.01a	0.19 ± 0ab
Fe (mg/L)	2.38 ± 0.12b	2.74 ± 0.12a	1.95 ± 0.17c	1.76 ± 0.12c	2.31 ± 0.11b	1.79 ± 0.14c	2.36 ± 0.14b	1.91 ± 0.08c
TS (%)	1.74 ± 0.02a	1.64 ± 0.01 cd	1.61 ± 0.01d	1.69 ± 0.01b	1.61 ± 0.01d	1.52 ± 0.01e	1.67 ± 0.01bc	1.69 ± 0.03b
TVS (g/L)	11.28 ± 0.07a	10.35 ± 0.14c	10.34 ± 0.10c	10.54 ± 0.06bc	10.21 ± 0.16c	9.73 ± 0.12d	10.33 ± 0.15c	10.78 ± 0.26b

Note: VFA/TA: ratio of the VFA and TA (g HAc equivalent to g $CaCO_3$ equivalent). Data are shown as mean ± standard error of duplicate reactors in each group with three replicate measurements for each parameter (n = 6). Data in each row with different letters ("a" - highest value and "f" - lowest value) indicate significant differences ($p < 0.05$).

confirming stable metabolic conditions and balanced acid production-consumption dynamics in all reactors despite varying

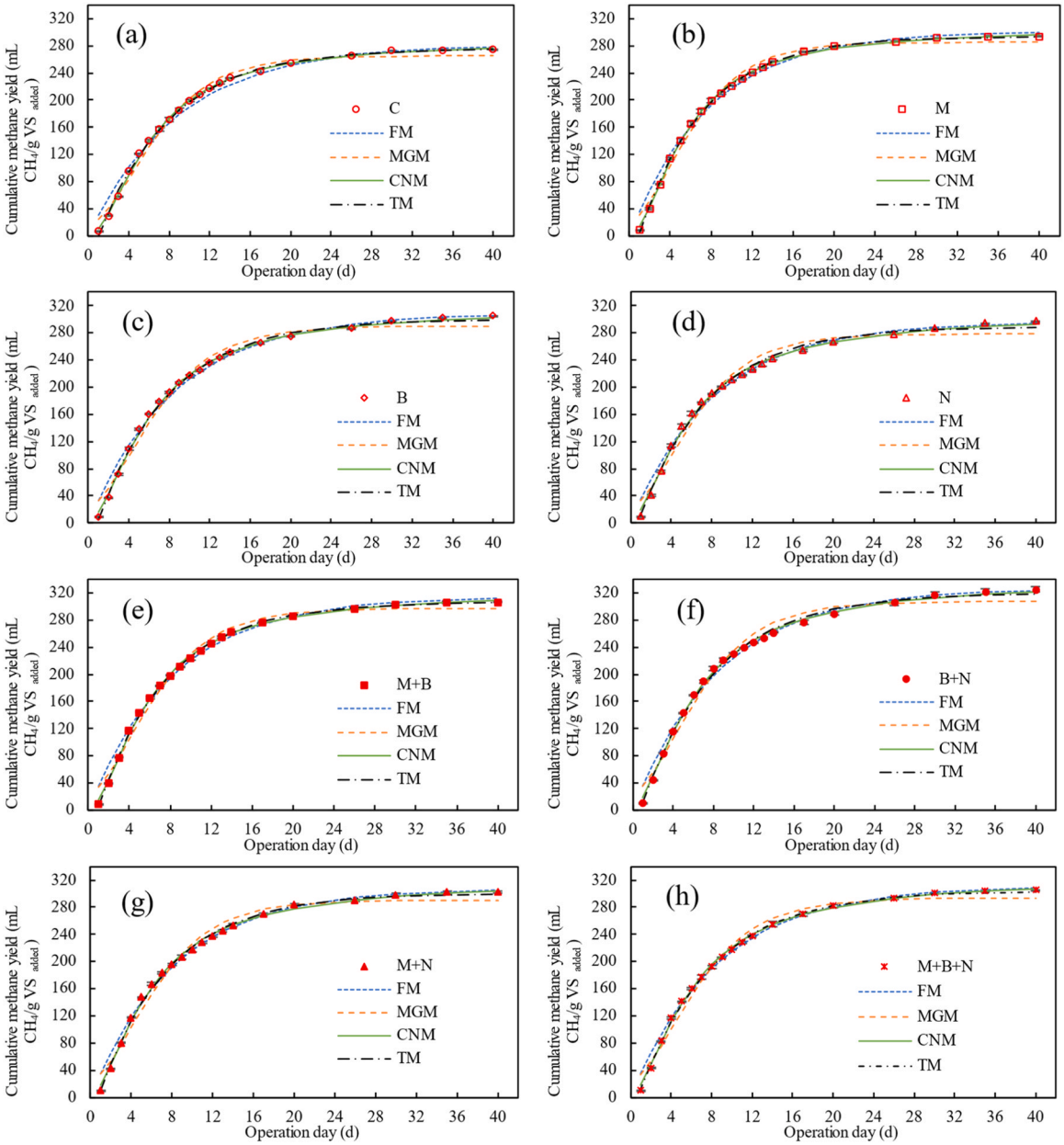


Fig. 2. The fitting of different kinetic models to the measured cumulative methane yield (NmL CH_4/g VS_{added}) from the Co-AD of PL and WS receiving different strategies (FM: first-order model; MGM: modified Gompertz model; CNM: cone model; TM: transfer model).

enhancement strategies. In addition, all enhancement treatments achieved lower final TS and TVS concentrations compared to the control, indicating improved organic matter degradation. The B+N treatment demonstrated the lowest residual solids levels (1.52 ± 0.01% TS and 9.73 ± 0.12 g/L TVS), corroborating the superior substrate hydrolysis and methanogenic conversion observed in this treatment, as evidenced by its highest SMY.

The substrate TS and TVS removal across different experimental groups are presented in Fig. 1**(c)**. All enhancement treatments achieved significantly improved removal rates relative to the control. The B+N exhibited the highest removal for both TS (80.2 ± 0.9%) and TVS (86.3 ± 1.3%), representing improvements of 38.6% and 24.2% over the control ($p < 0.05$), respectively. These results confirm that the synergistic combination of B and N enhanced the substrates' degradation most effectively, consistent with the superior SMY and favorable digestate quality parameters observed above. The enhanced degradation could be attributed to the complementary mechanisms of B-facilitated DIET and N-mediated trace element supplementation, which collectively optimized hydrolytic, acetogenic, and methanogenic activities. However, the M+B+N treatment demonstrated significantly lower TS and TVS removal compared to B+N ($p < 0.05$), indicating that M introduction compromised the substrate degradation efficiency achieved by the B+N. This reduction may result from oxygen-induced shifts in microbial community structure toward facultative aerobes that incompletely oxidize organic substrates or from disruption of the optimized anaerobic metabolic networks established by B and N. These findings reinforce the conclusion that M exhibits antagonistic rather than synergistic interactions with B-N amendments, further supporting the recommendation of B+N as the optimal enhancement strategy for this Co-AD process.

3.2. Kinetic study of methane yield

3.2.1. Kinetic modeling and evaluation

The model fitting curves for the four kinetic models applied to measured CMY data across all eight groups are presented in Fig. 2. The CNM and TM demonstrated superior goodness-of-fit compared to the FM and MGM across all groups, exhibiting higher R^2 values, lower RMSE, and lower AIC values. The evaluation parameters and kinetic constants for CNM and TM are presented in Table 4, while those for FM and MGM are provided in Table S1.

CNM achieved R^2 values of 0.9967–0.9990, RMSE of 2.67–4.67, and AIC of 49.89–72.33, while TM showed comparable performance with R^2 of 0.9950–0.9991, RMSE of 2.59–5.77, and AIC of 48.77–80.79. In contrast, FM and MGM exhibited lower R^2 values (0.9762–0.9910), higher RMSE (8.14–12.55), and higher AIC (91.37–111.87). The fitting curves in Fig. 2 visually confirm the superior correlation between CNM/TM predictions and experimental data compared to FM/MGM. These results demonstrate that CNM and TM provide more accurate simulations of methane yield kinetics in the Co-AD process across all enhancement strategies.

3.2.2. Kinetic constants

The kinetic constants obtained through model fitting provided a quantitative characterization of methane production dynamics under different enhancement strategies. These parameters included ultimate specific methane yield (Y_m, NmL CH_4/g VS_{added}) determined by all models, hydrolysis rate constant (k, day^{-1}) from FM and CNM, and maximum methane production rate (Rm, NmL CH_4/g VS_{added}/day) and lag phase (λ, day) from MGM and TM (Table 4). Based on the superior predictive accuracy of CNM and TM, kinetic parameters from these models were employed for comparative analysis of treatment effects.

a. The specific methane yield of substrate (Y_m)

The predicted final specific methane yield (Y_m, NmL CH_4/g VS_{added}) and corresponding deviation (Dif %) from the observed CMY values are presented in Table 4. Y_m serves as a predictor of methane production potential based on substrate utilization [52]. The TM provided Y_m predictions closer to the observed CMY values with lower absolute deviations compared to CNM, though TM systematically underpredicted Y_m in all treatment groups (negative Dif %, indicating the lower values of the predicted Y_m compared to the measured CMY values). The Y_m values predicted by both CNM and TM followed similar trends to the observed CMY, with B+N achieving the highest values (339.89 and 319.12 NmL CH_4/g VS_{added} for CNM and TM, respectively), followed by B, M+B, M+N, and M+B+N, then N and M, all exceeding the control. These results confirm that kinetic modeling accurately captured the enhanced substrate utilization and methane production potential observed experimentally.

b. The hydrolytic rate constant (k)

The hydrolysis rate constant (k, day^{-1}), derived from FM and CNM,

Table 4
Summary of the kinetic parameters using the cone model (CNM) and the transfer model (TM) for the Co-AD process receiving different strategies.

Groups		C	M	B	N	M+B	B+N	M+N	M+B+N
Experimental CMY_o (NmL/g VS_{added})		274.43 ± 0.87	293.92 ± 1.12	304.66 ± 0.43	296.88 ± 0.68	306.61 ± 1.15	324.83 ± 3.78	302.58 ± 2.05	305.28 ± 0.33
CNM	Y_m (NmL/g VS_{added})	286.39	305.10	316.33	308.08	320.87	318.46	339.89	324.47
	Dif (%)	4.36	3.80	3.83	3.77	4.65	5.25	4.64	6.29
	k (day^{-1})	0.160	0.180	0.164	0.169	0.171	0.170	0.162	0.163
	n	1.77	1.76	1.64	1.54	1.68	1.58	1.55	1.55
	R^2	0.9989	0.9990	0.9983	0.9967	0.9986	0.9977	0.9982	0.9984
	RMSE	2.67	2.68	3.53	4.67	3.30	4.09	3.83	3.44
	AIC	49.89	50.13	61.08	72.33	58.41	67.02	64.40	60.03
TM	Y_m (NmL/g VS_{added})	275.70	293.89	300.20	289.16	306.28	300.41	319.12	303.91
	Dif (%)	0.46	-0.01	-1.47	-2.60	-0.11	-0.72	-1.76	-0.45
	R_m (NmL/g VS_{added}/day)	38.57	46.11	42.19	41.44	45.07	43.38	43.87	41.99
	λ (day)	1.01	0.92	0.86	0.72	0.87	0.74	0.74	0.73
	R^2	0.9982	0.9991	0.9975	0.9950	0.9989	0.9976	0.9969	0.9988
	RMSE	3.41	2.59	4.27	5.77	2.88	4.17	4.99	2.91
	AIC	59.76	48.77	68.75	80.79	52.92	67.77	74.96	53.44

Note: CMY_o = the actual observed cumulative methane yield (NmL CH_4/g VS_{added}); Y_m = specific methane yield of substrates (NmL CH_4/g VS_{added}); e = 2.718; k = hydrolytic rate constant (day^{-1}); t = digestion time (days); R_m = maximum methane production rate (NmL CH_4/g VS_{added}/day); λ = lag phase (days); n = shape constant, which reveals whether there is a lag phase in the reactor.
a. The specific methane yield of substrates (Y_m)

characterizes the substrate degradation rate, with higher values indicating faster degradation [53]. While CNM consistently yielded higher k values than FM, both models exhibited similar trends across the treatment groups. In CNM, all enhancement strategies increased k relative to the control (0.160 day^{-1}), demonstrating that all treatments facilitated substrate hydrolysis in the Co-AD process. M achieved the highest k value (0.180 day^{-1}, +12.0%), followed by M+B (0.171 day^{-1}), M+N (0.170 day^{-1}), N (0.169 day^{-1}), and B (0.164 day^{-1}), indicating M exerted the strongest effect on enhancing the hydrolysis rate. However, the lower k values in combined treatments, M+B and M+N, compared to M alone, suggest potential antagonistic interactions between M and either B or N. Furthermore, M+B+N and B+N exhibited the smallest k increases (0.163 and 0.162 day^{-1}, respectively), lower than any single-strategy treatment, indicating that combining B and N might inhibit hydrolysis rate enhancement despite their synergistic effects on Y_m.

c. Maximum methane production rate (R_m) and lag phase (λ)

R_m (NmL CH_4/g VS_{added}/day) and λ (day) are critical kinetic parameters characterizing methanogenic activity, where higher R_m and shorter λ indicate enhanced performance. Values obtained from TM revealed that all enhancement strategies increased R_m and reduced λ compared to the control (38.57 NmL CH_4/g VS_{added}/day and 1.01 day, respectively), confirming improved methanogenic activity across all treatments. M achieved the highest R_m (46.11 NmL CH_4/g VS_{added}/day, +19.5%), followed by M+B (45.07 NmL CH_4/g VS_{added}/day, +16.8%) and M+N (43.38 NmL CH_4/g VS_{added}/day, +12.5%), demonstrating M's dominant effect on enhancing the maximum production rate. The lower R_m values in M+B and M+N compared to M alone suggest antagonistic interactions between microaeration and the other amendments. Notably, B+N exhibited higher R_m (43.87 NmL CH_4/g VS_{added}/day) than either B or N individually (42.19 and 41.44 NmL CH_4/g VS_{added}/day, respectively), indicating synergistic enhancement. However, M+B+N showed decreased R_m (41.99 NmL CH_4/g VS_{added}/day), confirming that M negatively interacted with B+N, consistent with the SMY results mentioned above.

All treatments reduced λ relative to the control, with N-containing treatments (N, M+N, B+N, M+B+N) achieving the shortest λ values (0.72–0.74 days), followed by B treatments (B, M+B: 0.86–0.87 days), and M alone showing the least reduction (0.92 days). These results demonstrate that N most effectively shortened λ, followed by B and then M. The superior performance of N in reducing startup time likely reflects rapid trace element bioavailability that facilitates immediate methanogenic enzyme activation [54,55].

3.3. Microbial community shift

3.3.1. Overall phylum and genus community

The phylum-level microbial community structure exhibited general similarity across all treatment groups (Fig. 3 **(a)**), dominated by *Chloroflexi* (26–33% relative abundance), followed by an *unidentified_Bacteria* (11–15%), *Actinobacteria* (10–13%), *Cloacimonadota* (4.5–7.6%), *Bacteroidota* (5.6–8.4%), *Firmicutes* (6.0–7.2%), *Acidobacteriota* (4.8–7.1%), *Proteobacteria* (3.8–5.6%), *Desulfobacterota* (1.7–2.4%), and *Halobacterota* (0.6–1.6%).

Relative to the initial sludge (O), *Cloeochromatiota* declined from 10.0%, whereas *unidentified_Bacteria*, *Firmicutes*, *Acidobacteriota*, and *Desulfobacterota* increased from 9.8%, 3.3%, 4.2%, and 1.3%, respectively, demonstrating selective microbial enrichment. Compared to the control (C), phylum-level abundances exhibited treatment-specific variations, with certain phyla enriched in some groups while depleted in others, and vice versa. Except for the *unidentified Bacteria* and *Halobacterota*, all other phyla were well recognized as common hydrolytic and fermentative microorganisms that participated in the breakdown of organic matter during the AD process [56]. *Euryarchaeota*, an archaeal phylum predominantly comprising methanogenic organisms, including hydrogenotrophic and acetoclastic methanogens, represented 0.1% in O and increased to 0.3–0.4% in C and treatment groups.

At the genus level (Fig. 3 **(b)**), *unidentified_SBR1031* maintained consistent dominance (10–15% relative abundance) across all treatments, indicating its functional importance and resilience to operational

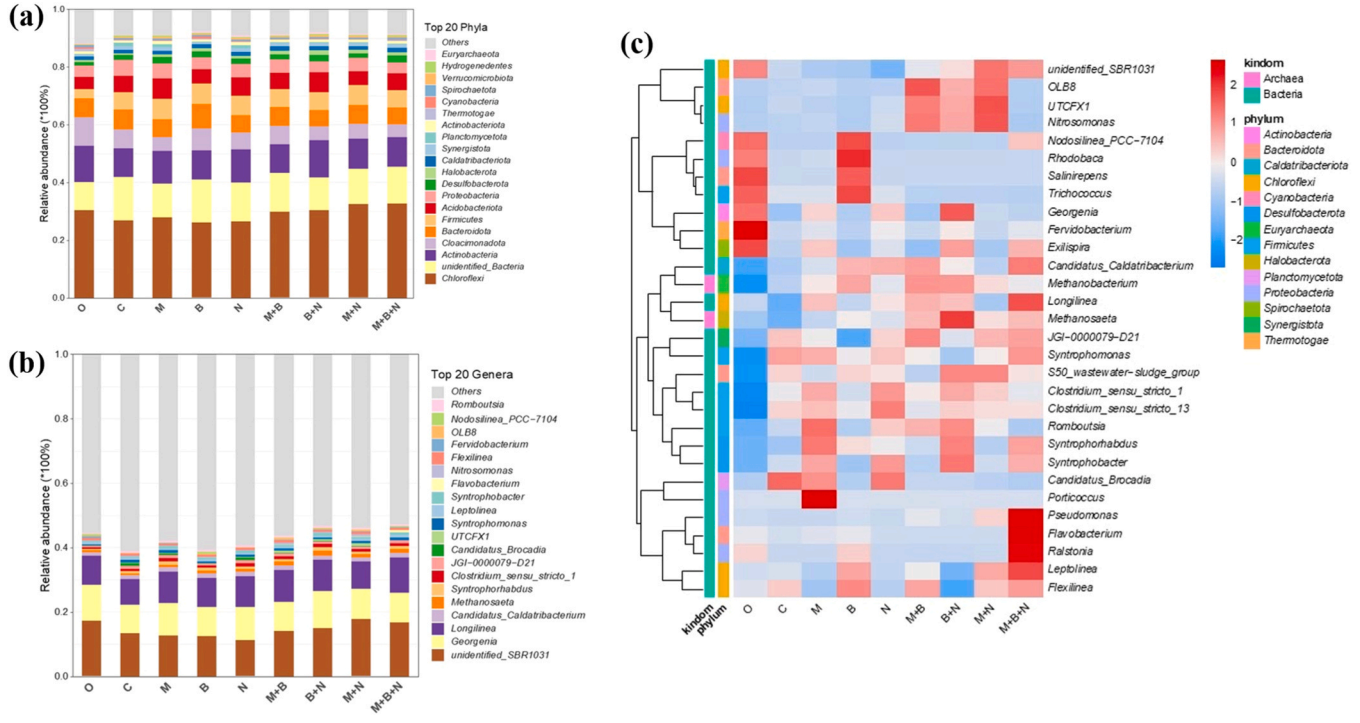


Fig. 3. Microbial community structure across treatment groups: relative abundance of (a) the top 20 phyla, (b) the top 20 genera, and (c) the genus-level abundance heatmap of the top 30 genera scaled by the row in sludge samples of the Co-AD process.

perturbations. Previous studies have identified *SBR1031* as a heterotrophic bacterium with metabolic functions associated with ammonia nitrogen metabolism, particularly the degradation of nitrogen-sulfur compounds in petroleum-derived substrates during AD [57]. *Longilinea,* known as a filamentous bacterium that produces acetate, lactate, and hydrogen as final products using fructose, sucrose, raffinose, or xylose under anaerobic conditions [58], showed enriched abundance across all treatment groups (8.6–10.9%), compared to the C (8.0%). The hydrolytic bacteria *Candidatus_Caldatribacterium* [59] also showed substantial representation across all treatment samples (1.2–1.5%). Critically, the single and combined enhancement treatments (M, B, N, M+B, B+N, M+N, M+B+N) enhanced the relative abundance of *Methanosaeta* (0.7–1.5%) compared to C (0.5%), with B+N showing the most pronounced enrichment (1.5% of total relative abundance). *Methanosaeta* is known as a key acetoclastic methanogen responsible for methane production from acetate [59]. A genus-level abundance heatmap was constructed to clearly illustrate community differences across treatment groups, as discussed in detail in the following subsection.

3.3.2. Heatmap analysis

The hierarchical clustering heatmap of the top 30 genera was constructed (Fig. 3 **(c)**) to reveal distinct treatment-specific microbial enrichment patterns relative to O and C. The dendrogram clearly separated samples into major clusters, and the color gradient showed the abundance difference of each genus in different groups. The acetoclastic *Methanosaeta* and the hydrogenotrophic *Methanobacterium* were the two archaeal genera, while the other 28 genera were bacterial genera. Overall, the initial sludge (O) showed a distinct community profile compared with all digested samples, while individual and combined treatments imposed differentiated selective pressures on both bacterial and archaeal genera.

Compared to O, C showed moderate shifts in genus-level composition, but remained closer to O than most amended treatments. Most taxa persist in C, except that some fermentative genera decreased, including the heterotrophic bacteria *unidentified_SBR1031*, the redox-responsive and facultative genera *Georgenia*, *Flavobacterium*, *Nodosilinea_PCC-7104, Romboutsia, Exilispira, Salinirepens, Rhodobaca,* and *Trichococcus,* while the *Candidatus_Brocadia* increased.

Among the single-treatment groups, M exhibited the highest relative abundance of *Porticoccus* compared with all other groups, while also displaying a relatively even and broadly enriched community composition. B showed notable increases in several symbiotic and facultative genera, including *Nodosilinea_PCC-7104*, *Salinirepens*, and *Rhodobaca* [58], as well as in the fermentative genus *Trichococcus* that was reported to play a key role in starch degradation to acetate [60]. In contrast, N showed a more evenly distributed genus composition, with the dominant genera showing slight relative increases rather than strong enrichment of specific taxa.

The combined treatments exhibited more pronounced and genus-specific redistribution patterns than single amendments. In the M+B group, the genera were distributed similarly to B except for a shift of increased genera from *Nodosilinea_PCC-7104*, *Salinirepens*, and *Rhodobaca* to *OLB8*, *UTCFX1*, and *Nitrosomonas*, and the increased abundance of *Methanosaeta* and *Methanobacterium*. In the B+N group, the facultative genera of *Georgenia*, known as an efficient azo dye decolorizing bacterium [60] and the syntrophic bacteria of *Syntrophorhabdus* and *Syntrophobacter*, being essential for degrading VFAs in cooperation with hydrogenotrophic methanogens [61], were both elevated in relative abundance, compared to C. The genus abundance patterns in the M+N group were quite similar to M+B. The increase of *Nitrosomonas* in both M+N and M+B groups indicated that M selectively favored this aerobic microorganism related to nitrification and denitrification [62]. M+B+N showed pronounced enrichment (intense red coloration) of *Longilinea*, *Pseudomonas*, *Flavobacterium*, *Ralstonia*, and *Leptolinea*, which are aerobic or facultative genera known for metal resistance and extracellular electron transfer capabilities [61,63,64].

In addition, *Methanosaeta* exhibited a progressive increase in relative abundance from the C to the single-treatment groups and further to the combined treatments (M+B, M+N, and M+B+N), reaching its highest level in the B+N group (dark red). *Methanobacterium* also increased across all treatment groups compared with C, with more pronounced enrichment observed in the B-amended groups (B, M+B, and B+N). The clear separation between C and enhanced treatments demonstrated that enhancement strategies fundamentally altered community structure through multiple mechanisms, as mentioned previously. These findings indicated that strategic enhancement combinations created functionally diverse microbial communities that substantially improved Co-AD system performance. The synergistic effects observed in the combined treatments suggest optimization of syntrophic relationships between fermentative bacteria, syntrophic acetogens, and methanogenic archaea, thereby potentially enhancing the overall efficiency and stability of the Co-AD process.

The heatmap analysis further supports the differences between B+N and M+B+N. The B+N group showed relatively higher abundance of key methanogens (e.g., *Methanosaeta* and *Methanobacterium*) and syntrophic/electroactive bacteria (e.g., *Syntrophobacter*), which are associated with efficient methane production and electron transfer. In contrast, the M+B+N group exhibited a shift toward a more dispersed community with relatively lower enrichment of these functional taxa and increased abundance of other genera less directly linked to methanogenesis. These patterns suggest that M may alter microbial interactions and weaken the syntrophic relationships established in B+N, which is consistent with the observed reduction in methane production, although the mechanism remains indicative rather than conclusive.

3.3.3. Archaeal community and predictive methanogenic pathways analysis

Archaea, though representing a minor fraction of the total community (approximately 0.8–2.0% relative abundance), exhibited substantial compositional and functional variations in response to the enhancement strategies (Fig. 4 **(a)**). The sludge at the initial sludge (O) and the control (C) harbored relatively low archaeal abundances (approximately 0.8–0.9%), dominated by *Methanosaeta* and *Methanobacterium*, representing the baseline methanogenic capacity. Single-amendment treatments (M, B, N) showed moderate increases in total archaeal abundance (1.0–1.5%), with M and B particularly enriching *Methanosaeta*, consistent with the genus-level patterns observed in Fig. 3 **(b)**. Notably, the combined enhancement treatments (M+B, B+N, M+N, M+B+N) achieved the highest archaeal abundances (1.5–2.0%), with M+B and B+N treatments showing the most substantial enrichment. The dominance of *Methanosaeta* across all enhanced treatments, which increased progressively from single to combined amendments, confirms the preferential stimulation of acetoclastic methanogenesis, which is the primary methane production pathway in AD systems [58,59,65]. *Methanobacterium*, a hydrogenotrophic methanogen, maintained relatively stable abundance across treatments, while *Candidatus_Methanofastidiosum* and minor genera such as *Methanoculleus* and *Candidatus_Methanomethylicus* showed slight enrichment in specific combined treatments, suggesting diversification of methanogenic pathways [65–67].

The functional prediction based on KEGG database analysis substantiated these community shifts, revealing that methanogenic pathway abundance increased correspondingly with archaeal enrichment (Fig. 4 **(b)**). The METHANOGENESIS-PWY (acetoclastic pathway) and METH-ACETATE-PWY pathways dominated across all samples, comprising over 99.9% of predicted methanogenic functions, with subtle but discernible increases in combined treatments. The methylotrophic pathway (PWY-6830) remained minor but showed slight elevation in N-amended treatments, possibly reflecting the emergence of methylotrophic methanogens under altered redox conditions [61,63]. These results mechanistically explain the performance improvements suggested by community restructuring in Fig. 3, i.e., M likely enhances substrate oxidation efficiency, making more acetate available for

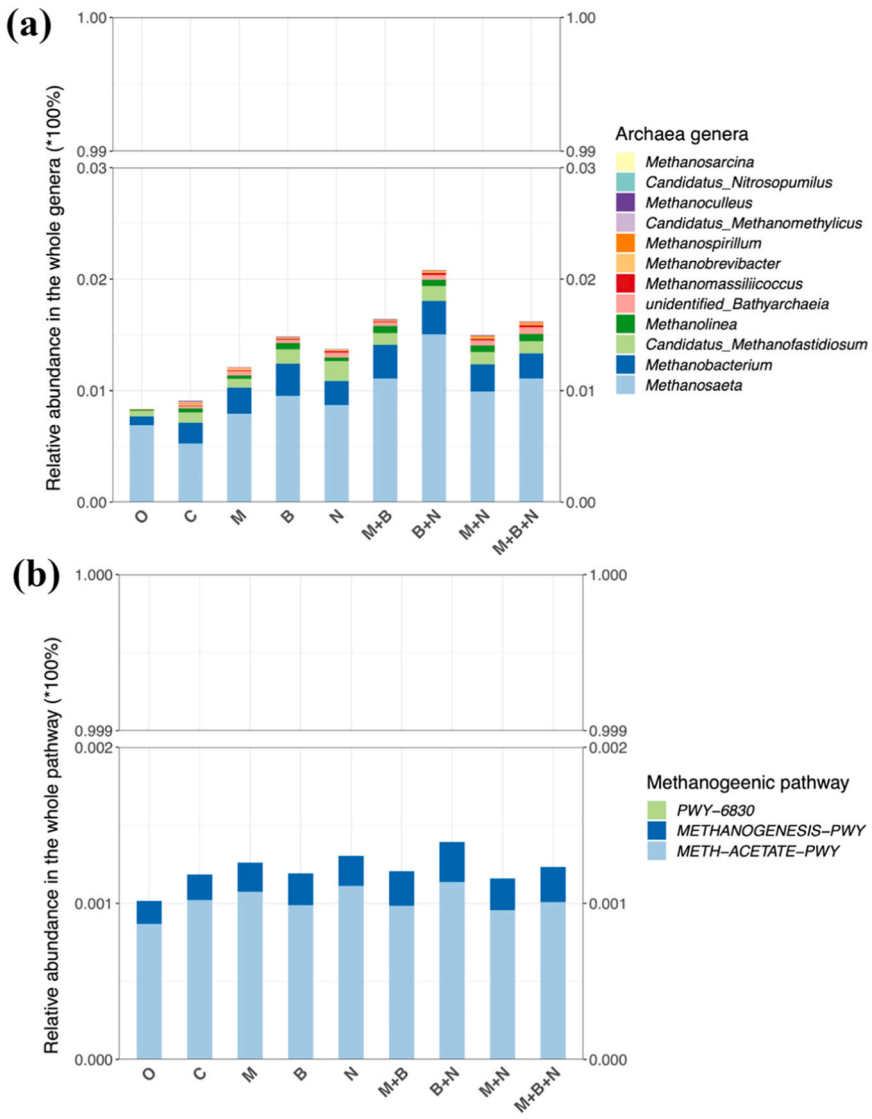


Fig. 4. (a) Relative abundance of archaea genera in the entire community in different groups and (b) Relative abundance of methanogenic pathways inferred from the Kyoto Encyclopedia of Genes and Genomes (KEGG) database by functional prediction.

Methanosaeta; B facilitates DIET between syntrophic bacteria (*Syntrophomonas*, *Syntrophobacter*) and methanogens, creating favorable microenvironments for archaeal colonization; and N provides essential trace elements (iron) that serve as cofactors for methanogenic enzymes. The synergistic enrichment of both syntrophic bacteria (Fig. 3) and acetoclastic methanogens under combined treatments establishes a robust syntrophic network wherein VFAs produced by fermentative bacteria are efficiently converted to acetate by syntrophs and subsequently to methane by *Methanosaeta*, optimizing the overall methanogenic flux. The relative stability of hydrogenotrophic methanogenesis across treatments, despite substantial variations in acetoclastic capacity, suggests that hydrogen-mediated methanogenesis operated near saturation in all systems, whereas acetoclastic methanogenesis remains the rate-limiting step that responds most dramatically to the enhancement strategies.

This functional specialization explains why combined treatments, particularly B+N, achieved superior performance by simultaneously addressing multiple bottlenecks: enhancing substrate degradation, facilitating DIET, and enriching key methanogenic populations, thereby creating an integrated microbial consortium for efficient organic waste conversion to methane.

3.4. Cost-benefit analysis

The estimates of the costs and benefits from the energy values of the batch reactors and the assumed continuous ASBR in each group are summarized in Table 5. The net benefit value can be calculated by subtracting the added cost from the total estimated value of energy production. Therefore, the increased values of the treatment groups with enhancing techniques compared to C were used to demonstrate whether the enhanced energy production could make up for the added cost from those techniques. In the batch-scale experiments (200 mL working volume, 1.85 g VS), all enhancement strategies improved SMY relative to the control (274.4 NmL/g VS), with increases ranging from 7.1% for M alone (293.9 NmL/g VS) to 18.4% for the B+N (324.8 NmL/g VS). The corresponding E_p and V_p followed similar trends, with the B+N achieving the highest energy value (1.1×10^{-4} $/g VS) among all groups. When scaled up to a continuous ASBR framework (16 L working volume, 3.8 g VS/L_reactor/day OLR), the enhanced methane production translated to substantial increases in total daily energy production. B+N achieved the highest methane production (19.7 L/day) and energy value (6.6×10^{-3} $/day), representing an 18% improvement in volumetric methane production compared to C.

However, the economic viability of these enhancement strategies

Table 5
The cost and energy benefit analysis of batch Co-AD process enhanced by different techinuqes and the simulated calculation based on a continuous anaerobic sequencing batch reactor (ASBR) framwork.

Scienario	Batch reactor (200 mL, 1.85 g VS)				Continuous ASBR (16 L, 3.8 g VS/$L_{reactor}$/day)				
Group	Air (mL) or additives (mg)	SMY (NmL/g VS)	E_p (kWh/g VS)	V_p ($/g VS)	Air (mL/d) or additives (g)	Added cost total ($)	MP (L/d)	$E_{p\text{-}total}$ (kWh/d)	$V_{p\text{-}total}$ ($/d)
C	-	274.4	2.6×10^{-3}	9.2×10^{-5}	-	-	16.7	0.16	5.6×10^{-3}
M	4.26 mL	293.9	2.8×10^{-3}	9.9×10^{-5}	140.2 mL/d	$n \times 2.4 \times 10^{-4}$	17.9	0.17	6.0×10^{-3}
B	213 mg	304.7	2.9×10^{-3}	1.0×10^{-4}	17.04 g	0.051	18.5	0.18	6.2×10^{-3}
N	4.26 mg	296.9	2.8×10^{-3}	1.0×10^{-4}	0.34 g	0.034	18.1	0.17	6.1×10^{-3}
M+B	4.26 mL M + 213 mg B	306.6	2.9×10^{-3}	1.0×10^{-4}	17.04 g B + 140.2 mL/d	$n \times 2.4 \times 10^{-4}$ + 0.051	18.6	0.18	6.3×10^{-3}
B+N	213 mg B + 4.26 mg N	324.8	3.1×10^{-3}	1.1×10^{-4}	17.04 g B + 0.34 g N	0.085	19.7	0.19	6.6×10^{-3}
M+N	4.26 mL M + 4.26 mg N	302.6	2.9×10^{-3}	1.0×10^{-4}	0.34 g N + 140.2 mL/d	$n \times 2.4 \times 10^{-4}$ + 0.034	18.4	0.18	6.2×10^{-3}
M+B+N	4.26 mL M + 213 mg B + 4.26 mg N	305.3	2.9×10^{-3}	1.0×10^{-4}	17.04 g B + 0.34 g N + 140.2 mL/d	$n \times 2.4 \times 10^{-4}$ + 0.085	18.6	0.18	6.2×10^{-3}

Note: SMY: Specific methane yield (NmL/g VS); E_p: Estimated energy production on VS basis (KJ/g VS); V_p: estimated energy value on VS basis (($/g VS); Added cost is the cost of added enhancing technogies, i.e., air pump microaeration, biochar, and Fe_3O_4 nanoparticles; MP: methane production (L); $E_{p\text{-}total}$: total estimated energy production (KJ); $V_{p\text{-}total}$: total estimated energy value ($).

critically depends on the operational cost of the additives and the system's operational lifespan. M incurred negligible costs (≈2.4 $\times 10^{-4}$ $/day for air pumping), making it highly cost-effective despite its moderate performance improvement. B required a substantial upfront investment (0.051 $/day based on continuous dosing at 17.04 g/day), while N showed intermediate costs (0.034 $/day at 0.34 g/day). The combined treatments exhibited additive cost structures, with B+N showing the highest operational cost (0.085 $/day) and M+B+N reaching similar levels. The net increased benefit analysis revealed distinct economic trajectories for different enhancement strategies as a function of operational time (Fig. S3). Initially, all enhancement treatments showed a negative increase in net benefits due to upfront investment costs, with the triple amendment (M+B+N) exhibiting the most pronounced deficit (approximately -$0.08) at day 30. However, as operational time extended, the cumulative energy gains progressively offset the added costs. The B+N treatment demonstrated the steepest positive trajectory, reaching break-even around day 120 and achieving the highest net benefit ($0.29) by day 360, attributable to its superior methane production enhancement combined with the absence of continuous biochar replenishment costs in this scenario. B showed the second-best long-term economic performance, reaching approximately $0.17 net benefit at day 360, while N achieved $0.13. M treatments (M, M+N, M+B) clustered together with modest net benefits ($0.04–0.08 by day 360), as their lower operational costs were offset by more moderate performance improvements. Notably, the triple amendment (M+B+N) showed recovery from its initial deficit but remained below B+N and B alone, suggesting that the incremental benefit of combining all three strategies did not justify the cumulative costs. This outcome aligns with the microbial community analysis (Figs. 3–4**)**, where M+B+N showed similar community structure and archaeal enrichment to B+N, indicating redundancy in enhancement mechanisms.

The break-even analysis reveals critical decision points for practical implementation, i.e., M becomes economically favorable almost immediately due to negligible costs, B requires approximately 90–120 days of operation to recover costs, and N shows intermediate payback periods of 60–90 days. For short-term operations (<180 days), M or N amendments offer the best cost-effectiveness, while for long-term installations (>180 days), B+N provides superior net economic benefits. The diverging trajectories after day 180 (Fig. S3) underscore the importance of operational planning, i.e., industrial-scale continuous digesters with multi-year operational horizons would benefit most from B+N implementation, potentially achieving net benefits exceeding $100–200 annually per cubic meter of reactor volume, whereas pilot-scale or seasonal operations might favor low-cost M strategies.

3.5. Limitations and perspectives

Microbial community analysis was performed without biological replication. Therefore, statistical comparisons among treatments were not conducted. As a result, subtle differences, particularly for low-abundance taxa, should be interpreted with caution, and the observed trends are presented as supportive rather than definitive evidence. Future studies incorporating biological replication and statistical analysis are recommended to validate the observed community shifts.

Although strong correlations between methanogen abundance and methane production were observed in this study, substrate conversion efficiency would depend not only on microbial abundance but also on functional gene expression and enzymatic activities. While the current microbial community analysis focused on taxonomic composition, future work should use metatranscriptomics and metaproteomics to directly assess functional activity. For example, functional gene profiling targeting key metabolic pathways would directly link community structure to metabolic capacity; co-occurrence network analysis and metabolite tracking could elucidate syntrophic interactions between community members. Spatial distribution analysis, particularly examining microbial colonization on biochar surfaces versus bulk liquid, would reveal microhabitat preferences.

The involvement of mechanisms such as DIET and M-enhanced hydrolysis in this study is inferred from indirect evidence, including process performance, kinetic parameters, and relative shifts in microbial community composition. While the enrichment of potential electroactive microorganisms and methanogens, together with improved methane production kinetics, may suggest conditions favorable for DIET, these observations are not sufficient for definitive confirmation, particularly given the overall similarity in community structures across treatments. Therefore, the proposed mechanisms should be considered plausible but not conclusive. Advanced analytical approaches, such as electrochemical characterization, functional gene analysis, or transcriptomics, are required to directly verify these processes in future studies.

Due to the low oxygen dosage and the high oxygen uptake capacity of anaerobic microorganisms and reduced intermediates, the introduced oxygen was expected to be rapidly consumed rather than accumulated in the system. However, dissolved oxygen (DO), oxidation-reduction potential (ORP), or gas-phase oxygen concentrations were not directly monitored in this study. Therefore, the oxygen dynamics could not be quantitatively assessed. Future work incorporating direct measurements of DO, ORP, or oxygen profiles is needed to characterize oxygen behavior better and validate the micro-aeration conditions. In addition, this study was conducted under dilute conditions with TS at 2.13%.

Industrial systems are typically operated at < 15% TS (wet AD) or > 15% TS (dry AD) [68], where mass transfer limitations, microbial dynamics, and potential ammonia inhibition differ substantially. Systematic evaluation across varying TS levels is needed to establish operational windows and optimize dosing strategies for different application scenarios. Batch reactors employed here have enabled fundamental performance assessment but do not capture challenges of continuous operation, including process stability under dynamic loading, shock load resilience, and long-term enhancement effects.

It should be noted that the techno-economic assessment was based on a simulated continuously operated ASBR system and is intended to provide a comparative evaluation rather than a precise prediction of full-scale performance. To ensure consistency, several simplifying assumptions were applied, including the single addition of B and N at startup, which was supported by biochar's recalcitrance and long-term stability as a microbial carrier [69] and the chemical stability of Fe_3O_4 nanoparticles [70]. While the magnetic properties of Fe_3O_4 suggest feasibility for recovery via magnetic separation, the recovery efficiency, material loss, and long-term stability were not experimentally validated. In practical continuous systems, both B and N may undergo degradation, attrition, or washout, requiring periodic replenishment, which would increase operational costs and potentially affect the relative economic performance of all amended systems. Therefore, the results should be interpreted with caution. Conversely, potential cost reductions through bulk purchasing of N or in-situ B production from waste biomass could further improve economic viability. Future studies incorporating experimental validation of material recovery, stability, and continuous operation data are needed to improve the robustness and practical relevance of the techno-economic assessment.

The findings of this study highlighted the importance of strategy selection and interaction effects in optimizing Co-AD process performance. While the B+N combination showed the most favorable results under the tested conditions, the observed antagonistic effect in the M+B+N system suggested that overlapping or competing mechanisms may limit synergistic benefits. Future studies should further investigate these interactions using advanced analytical techniques (e.g., electrochemical characterization and functional gene analysis) to better resolve underlying mechanisms. In addition, validation under continuous or pilot-scale systems, along with consideration of material stability, recovery, and long-term operation, will be critical to translating these findings into practical applications.

Besides, substrate-specific responses require further investigation, as lignocellulosic materials may exhibit different responses to M due to their recalcitrant structures, while lipid-rich substrates could benefit more from B's capacity to mitigate long-chain VFA inhibition. Comprehensive evaluation across diverse substrate portfolios would establish the generalizability of these enhancement strategies and help develop substrate-specific optimization protocols. Future research priorities should include multi-factorial optimization approaches employing response surface methodology or machine learning algorithms, investigation of functionalized materials such as magnetic biochar and alternative nanoparticles, and integration with complementary pretreatment technologies (thermal, enzymatic). Finally, pilot-scale validation to comprehensively assess methane production, digestate quality, greenhouse gas emissions, and environmental impacts is essential to ensure holistic sustainability and facilitate technology transfer to full-scale industrial implementation.

4. Conclusions

This study systematically evaluated M, B, and N amendments for enhancing the Co-AD of PL with WS. All strategies improved SMY compared to the control (274.4 NmL CH_4/g VS), with B+N achieving the highest performance (324.8 ± 3.8 NmL CH_4/g VS, +18.4%). B+N also achieved superior solids removal (80.2% TS, 86.3% VS). Kinetic modeling revealed distinct mechanisms, i.e., M increased k by 12.0% and R_m by 19.5%, B enhanced Y_m, while N shortened λ to 0.72–0.74 days. Microbial community analysis demonstrated that all strategies enriched the acetoclastic methanogen *Methanosaeta*, with B+N achieving the highest abundance. Mechanistically, M enriched hydrolytic bacteria enhancing substrate degradation, B facilitated DIET and microbial colonization while enriching syntrophic bacteria, and N supplied trace elements and promoted DIET. Cost-benefit analysis showed that B+N achieved the highest long-term net benefit ($0.29 after 360 days) with break-even at 90–120 days, making it optimal for long-term installations, while M was suited for short-term operations. Overall, B+N is recommended as the optimal enhancement strategy. This study provides mechanistic insights into synergistic microbial-material interactions and establishes an integrated framework for process optimization to pave the way for industrial-scale AD proliferation. Future validation under diverse operational conditions, multi-omics investigations, and pilot-scale sustainability assessments are necessary for advancing the circular bioeconomy through improved waste valorization.

CRediT authorship contribution statement

Yuanhang Zhan: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Yiting Xiao:** Resources, Methodology, Investigation. **Sarah Wu:** Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Zhiwu Wang:** Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Jun Zhu:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This project is funded by USDA/NIFA/AFRI Applied Science and Foundational Program (Grant#: 2019–67021–29945), and the authors want to show appreciation for the financial support provided by the United States Department of Agriculture.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jece.2026.123307.

Data availability

Data will be made available on request.

References

[1] P. Abdeshahian, J.S. Lim, W.S. Ho, H. Hashim, C.T. Lee, Potential of biogas production from farm animal waste in Malaysia, Renew. Sustain. Energy Rev. 60 (2016) 714–723, https://doi.org/10.1016/j.rser.2016.01.117.
[2] M. Khalil, M.A. Berawi, R. Heryanto, A. Rizalie, Waste to energy technology: The potential of sustainable biogas production from animal waste in Indonesia, Renew. Sustain. Energy Rev. 105 (2019) 323–331, https://doi.org/10.1016/j.rser.2019.02.011.
[3] S. Xie, F.I. Hai, X. Zhan, W. Guo, H.H. Ngo, W.E. Price, et al., Anaerobic co-digestion: A critical review of mathematical modelling for performance optimization, Bioresour. Technol. 222 (2016) 498–512, https://doi.org/10.1016/j.biortech.2016.10.015.
[4] R. Karki, W. Chuenchart, K.C. Surendra, S. Shrestha, L. Raskin, S. Sung, et al., Anaerobic co-digestion: Current status and perspectives, Bioresour. Technol. 330 (2021) 125001, https://doi.org/10.1016/j.biortech.2021.125001.

[5] X. Zhu, D. Yellezuome, R. Liu, Z. Wang, X. Liu, Effects of co-digestion of food waste, corn straw and chicken manure in two-stage anaerobic digestion on trace element bioavailability and microbial community composition, Bioresour. Technol. 346 (2021) 126625, https://doi.org/10.1016/j.biortech.2021.126625.
[6] D. Botheju, B. Lie, R. Bakke, xygen Effects in Anaerobic Digestion - II, Model. Identif. Control. A Nor. Res. Bull. 31 (2) (2010) 55–65, https://doi.org/10.4173/mic.2010.2.2.
[7] H. Xu, Y. Li, D. Hua, Y. Zhao, L. Chen, L. Zhou, et al., Effect of microaerobic microbial pretreatment on anaerobic digestion of a lignocellulosic substrate under controlled pH conditions, Bioresour. Technol. 328 (2021) 124852, https://doi.org/10.1016/j.biortech.2021.124852.
[8] D. Nguyen, Z. Wu, S. Shrestha, P.H. Lee, L. Raskin, S.K. Khanal, Intermittent micro-aeration: New strategy to control volatile fatty acid accumulation in high organic loading anaerobic digestion, Water Res 166 (2019) 115080, https://doi.org/10.1016/j.watres.2019.115080.
[9] J.W. Lim, J.Y. Wang, Enhanced hydrolysis and methane yield by applying microaeration pretreatment to the anaerobic co-digestion of brown water and food waste, Waste Manag 33 (4) (2013) 813–819, https://doi.org/10.1016/j.wasman.2012.11.013.
[10] S.O. Masebinu, E.T. Akinlabi, E. Muzenda, A.O. Aboyade, A review of biochar properties and their roles in mitigating challenges with anaerobic digestion, Renew. Sustain. Energy Rev. 103 (2019) 291–307, https://doi.org/10.1016/j.rser.2018.12.048.
[11] P.R. Yaashikaa, P.S. Kumar, S. Varjani, A. Saravanan, A critical review on the biochar production techniques, characterization, stability and applications for circular bioeconomy, Biotechnol. Rep. 28 (2020) e00570, https://doi.org/10.1016/j.btre.2020.e00570.
[12] G.S. Ghodake, S.K. Shinde, A.A. Kadam, R.G. Saratale, G.D. Saratale, M. Kumar, et al., Review on biomass feedstocks, pyrolysis mechanism and physicochemical properties of biochar: State-of-the-art framework to speed up vision of circular bioeconomy, J. Clean. Prod. 297 (2021) 126645, https://doi.org/10.1016/j.jclepro.2021.126645.
[13] K. Luo, Y. Pang, D. Wang, X. Li, L. Wang, M. Lei, et al., A critical review on the application of biochar in environmental pollution remediation: Role of persistent free radicals (PFRs), J. Environ. Sci. 108 (2021) 201–216, https://doi.org/10.1016/j.jes.2021.02.021.
[14] L. Qiu, Y.F. Deng, F. Wang, M. Davaritouchaee, Y.Q. Yao, A review on biochar-mediated anaerobic digestion with enhanced methane recovery, Renew. Sustain. Energy Rev. 115 (2019) 109373, https://doi.org/10.1016/j.rser.2019.109373.
[15] Q. Qi, C. Sun, J. Zhang, Y. He, Y. Wah Tong, Internal enhancement mechanism of biochar with graphene structure in anaerobic digestion: The bioavailability of trace elements and potential direct interspecies electron transfer, Chem. Eng. J. 406 (2021) 126833, https://doi.org/10.1016/j.cej.2020.126833.
[16] H. Baniamerian, P.G. Isfahani, P. Tsapekos, M. Alvarado-Morales, M. Shahrokhi, M. Vossoughi, et al., Application of nano-structured materials in anaerobic digestion: Current status and perspectives, Chemosphere 229 (2019) 188–199, https://doi.org/10.1016/j.chemosphere.2019.04.193.
[17] Y.-J. Lee, D.-J. Lee, Impact of adding metal nanoparticles on anaerobic digestion performance – A review, Bioresour. Technol. 292 (2019) 121926, https://doi.org/10.1016/j.biortech.2019.121926.
[18] M. Yazdani, M. Ebrahimi-Nik, A. Heidari, M.H. Abbaspour-Fard, Improvement of biogas production from slaughterhouse wastewater using biosynthesized iron nanoparticles from water treatment sludge, Renew. Energy 135 (2019) 496–501, https://doi.org/10.1016/j.renene.2018.12.019.
[19] B. Salehi, L. Wang, Critical Review on Nanomaterials for Enhancing Bioconversion and Bioremediation of Agricultural Wastes and Wastewater, Energ. J. 15 (2022) 5387, https://doi.org/10.3390/en15155387(firstpublished online).
[20] A. Hassanein, A. Naresh Kumar, S. Lansing, Impact of electro-conductive nanoparticles additives on anaerobic digestion performance - A review, Bioresour. Technol. 342 (2021) 126023, https://doi.org/10.1016/j.biortech.2021.126023.
[21] Z. Zhang, L. Guo, Y. Wang, Y. Zhao, Z. She, M. Gao, et al., Application of iron oxide (Fe3O4) nanoparticles during the two-stage anaerobic digestion with waste sludge: Impact on the biogas production and the substrate metabolism, Renew. Energy 146 (2020) 2724–2735, https://doi.org/10.1016/j.renene.2019.08.078.
[22] Y. Liu, Q. Xiao, X. Ye, C. Wang, Z. Jia, J. Du, et al., Effect of different charged Fe3O4 nanoparticles on methane production for anaerobic digestion of wheat straw, J. Clean. Prod. 328 (2021) 129655, https://doi.org/10.1016/j.jclepro.2021.129655.
[23] Y. Zhan, B. Zuo, X. Cao, Y. Xiao, J. Zhu, Biochar enhanced anaerobic co-digestion of poultry litter and wheat straw: Performance, microbial analysis, and multiple factors' interaction, Renew. Energy 231 (2024) 120907, https://doi.org/10.1016/j.renene.2024.120907.
[24] Y. Zhan, J. Zhu, Y. Xiao, L.C. Schrader, S. Xiao Wu, N. Aka Robinson Jr, et al., Employing micro-aeration in anaerobic digestion of poultry litter and wheat straw: Batch kinetics and continuous performance, Bioresour. Technol. 368 (2023) 128351, https://doi.org/10.1016/j.biortech.2022.128351.
[25] Y. Zhan, X. Cao, J. Zhu, Enhanced bioenergy production via operational optimization of anaerobic co-digestion of chicken litter and straw waste with magnetite nanoparticles, Energy 347 (2026) 140428, https://doi.org/10.1016/j.energy.2026.140428.
[26] A. Apha, Standard methods for the examination of water and wastewater, 20, American Public Health Association, Washington, DC, 1998.
[27] Q. Niu, W. Qiao, H. Qiang, T. Hojo, Y.-Y. Li, Mesophilic methane fermentation of chicken manure at a wide range of ammonia concentration: Stability, inhibition and recovery, Bioresour. Technol. 137 (2013) 358–367, https://doi.org/10.1016/j.biortech.2013.03.080.
[28] J. Shen, J. Zhu, Methane production in an upflow anaerobic biofilm digester from leachates derived from poultry litter at different organic loading rates and hydraulic retention times, J. Environ. Chem. Eng. 5 (5) (2017) 5124–5130, https://doi.org/10.1016/j.jece.2017.09.049.
[29] M.A. Olojede, O. Ogunkunle, N. Ahmed, D. A,Pham, Quality of optimized biogas yields from co-digestion of cattle dung with fresh mass of sunflower leaves, pawpaw and potato peels, Cogent Eng. 5 (1) (2018) 1538491, https://doi.org/10.1080/23311916.2018.1538491.
[30] P. Llabrés-Luengo, J. Mata-Alvarez, Kinetic Study of the Anaerobic Digestion of Straw-Pig Manure Mixtures, Biomass 14 (2) (1987).
[31] K. Li, R. Liu, C. Sun, Comparison of anaerobic digestion characteristics and kinetics of four livestock manures with different substrate concentrations, Bioresour. Technol. 198 (2015) 133–140, https://doi.org/10.1016/j.biortech.2015.08.151.
[32] Y. Zhang, Z. Yang, R. Xu, Y. Xiang, M. Jia, J. Hu, et al., Enhanced mesophilic anaerobic digestion of waste sludge with the iron nanoparticles addition and kinetic analysis, Sci. Total Environ. 683 (2019) 124–133, https://doi.org/10.1016/j.scitotenv.2019.05.214.
[33] J.-J. Lay, Y.-Y. Li, T. Noike, Interaction between homoacetogens and methanogens in lake sediments, J. Ferment. Bioeng. 86 (5) (1998) 467–471.
[34] G.K. Kafle, L. Chen, Comparison on batch anaerobic digestion of five different livestock manures and prediction of biochemical methane potential (BMP) using different statistical models, Waste Manag 48 (2016) 492–502, https://doi.org/10.1016/j.wasman.2015.10.021.
[35] Z. Zahan, M.Z. Othman, T.H. Muster, Anaerobic digestion/co-digestion kinetic potentials of different agro-industrial wastes: A comparative batch study for C/N optimisation, Waste Manag 71 (2018) 663–674, https://doi.org/10.1016/j.wasman.2017.08.014.
[36] H.M. El-Mashad, Kinetics of methane production from the codigestion of switchgrass and Spirulina platensis algae, Bioresour. Technol. 132 (2013) 305–312, https://doi.org/10.1016/j.biortech.2012.12.183.
[37] R. Karki, W. Chuenchart, K.C. Surendra, S. Sung, L. Raskin, S.K. Khanal, Anaerobic co-digestion of various organic wastes: Kinetic modeling and synergistic impact evaluation, Bioresour. Technol. 343 (2022) 126063, https://doi.org/10.1016/j.biortech.2021.126063.
[38] M. Wang, S.X. Tang, Z.L. Tan, Modeling in vitro gas production kinetics: Derivation of Logistic–Exponential (LE) equations and comparison of models, Anim. Feed Sci. Technol. 165 (3) (2011) 137–150, https://doi.org/10.1016/j.anifeedsci.2010.09.016.
[39] H. Motulsky, A. Christopoulos, Fitting models to biological data using linear and nonlinear regression: a practical guide to curve fitting, Oxford University Press, 2004.
[40] D.D. Nguyen, B.H. Jeon, J.H. Jeung, E.R. Rene, J.R. Banu, B. Ravindran, et al., Thermophilic anaerobic digestion of model organic wastes: Evaluation of biomethane production and multiple kinetic models analysis, Bioresour. Technol. 280 (2019) 269–276, https://doi.org/10.1016/j.biortech.2019.02.033.
[41] Y. Zhan, X. Cao, Y. Xiao, X. Wei, S. Wu, J. Zhu, Start-up of co-digestion of poultry litter and wheat straw in anaerobic sequencing batch reactor by gradually increasing organic loading rate: Methane production and microbial community analysis, Bioresour. Technol. 354 (2022) 127232, https://doi.org/10.1016/j.biortech.2022.127232.
[42] Y. Zhan, Y. Xiao, L.C. Schrader, J. Zhu, A float meter based system for self-regulated discharging and feeding in a laboratory semi-CSTR for anaerobic digestion of chicken litter, J. Asabe. 0 (0) (2022) 0, https://doi.org/10.13031/ja.14804.
[43] A.G.O. Paranhos, O.F.H. Adarme, G.F. Barreto, S.Q. Silva, S.F. Aquino, Methane production by co-digestion of poultry manure and lignocellulosic biomass: Kinetic and energy assessment, Bioresour. Technol. 300 (2020) 122588, https://doi.org/10.1016/j.biortech.2019.122588.
[44] M. Mainardis, S. Flaibani, F. Mazzolini, A. Peressotti, D. Goi, Techno-economic analysis of anaerobic digestion implementation in small Italian breweries and evaluation of biochar and granular activated carbon addition effect on methane yield, J. Environ. Chem. Eng. 7 (3) (2019) 103184, https://doi.org/10.1016/j.jece.2019.103184.
[45] A.E. Cioabla, I. Ionel, G.-A. Dumitrel, F. Popescu, Comparative study on factors affecting anaerobic digestion of agricultural vegetal residues, Biotechnol. Biofuels 5 (1) (2012) 39, https://doi.org/10.1186/1754-6834-5-39.
[46] R. Jia, Y.-C. Song, Z. An, K. Kim, C.-Y. Lee, B.-U. Bae, A new comprehensive indicator for monitoring anaerobic digestion: a principal component analysis approach. processes, Journal 12 (2024) 59, https://doi.org/10.3390/pr12010059(firstpublished online).
[47] T.G. Induchoodan, N. Choran, A.S. Kalamdhad, Domestic sewage as a sustainable freshwater substitute for enhanced anaerobic digestion of lignocellulosic biomass, Sci. Rep. 14 (1) (2024) 31964, https://doi.org/10.1038/s41598-024-83546-6.
[48] K.H. Hansen, I. Angelidaki, B.K. Ahring, ANAEROBIC DIGESTION OF SWINE MANURE: INHIBITION BY AMMONIA, Water Res. 32 (1) (1998) 5–12, https://doi.org/10.1016/S0043-1354(97)00201-7.
[49] O. Yenigün, B. Demirel, Ammonia inhibition in anaerobic digestion: A review, Process Biochem. 48 (5) (2013) 901–911, https://doi.org/10.1016/j.procbio.2013.04.012.
[50] F. Tang, J. Tian, N. Zhu, Y. Lin, H. Zheng, Z. Xu, et al., Dry anaerobic digestion of ammoniated straw: Performance and microbial characteristics, Bioresour. Technol. 351 (2022) 126952, https://doi.org/10.1016/j.biortech.2022.126952.
[51] S. Zahedi, R. Solera, J.L. García-Morales, H. Ennouri, D. Sales, Evaluation of the effect of glycerol supplementation on the anaerobic digestion of real municipal solid waste in batch mode, Fuel 193 (2017) 15–21, https://doi.org/10.1016/j.fuel.2016.12.024.

[52] B. Shamurad, N. Gray, E. Petropoulos, S. Tabraiz, E. Membere, P. Sallis, Predicting the effects of integrating mineral wastes in anaerobic digestion of OFMSW using first-order and Gompertz models from biomethane potential assays, Renew. Energy 152 (2020) 308–319, https://doi.org/10.1016/j.renene.2020.01.067.
[53] G. Zhen, X. Lu, T. Kobayashi, Y.-Y. Li, K. Xu, Y. Zhao, Mesophilic anaerobic co-digestion of waste activated sludge and Egeria densa: Performance assessment and kinetic analysis, Appl. Energy 148 (2015) 78–86, https://doi.org/10.1016/j.apenergy.2015.03.038.
[54] H. Mao, Q. Yu, Z. Zhao, Y. Zhang, New Insights into Nano Magnetite Enhancing Anaerobic Digestion: Regulating the Intracellular Electron Bifurcation and Storing Energy for Methanogens to Respond to Unfavorable Conditions, ACS ES&T Water 4 (2) (2024) 520–530, https://doi.org/10.1021/acsestwater.3c00586.
[55] R.B. Jeyakumar, G.S. Vincent, Recent Advances and Perspectives of Nanotechnology in Anaerobic Digestion: A New Paradigm towards Sludge Biodegradability, Sustain. J. 14 (2022) 7191, https://doi.org/10.3390/su14127191(firstpublished online).
[56] X. Peng, S. Zhang, L. Li, X. Zhao, Y. Ma, D. Shi, Long-term high-solids anaerobic digestion of food waste: Effects of ammonia on process performance and microbial community, Bioresour. Technol. 262 (2018) 148–158, https://doi.org/10.1016/j.biortech.2018.04.076.
[57] H. Li, T. Cai, Y. Gao, Q. Dai, X. Liu, X. Chen, et al., Long-term performance, microbial evolution and spatial microstructural characteristics of anammox granules in an upflow blanket filter (UBF) treating high-strength nitrogen wastewater, Bioresour. Technol. 367 (2023) 128206, https://doi.org/10.1016/j.biortech.2022.128206.
[58] O. Fernando Herrera Adarme, B. Eduardo Lobo Baêta, M. Cardoso Torres, F. Camilo Otalora Tapiero, L. Vinicius Alves Gurgel, S. de Queiroz Silva, et al., Biogas production by anaerobic co-digestion of sugarcane biorefinery byproducts: Comparative analyses of performance and microbial community in novel single- and two-stage systems, Bioresour. Technol. 354 (2022) 127185, https://doi.org/10.1016/j.biortech.2022.127185.
[59] J. Zhang, Q. Meng, R. Jiang, Z. Yang, Q. Zhang, W. Luo, et al., Enhancing methane production from distillers' grains by cattle manure in solid-state anaerobic digestion: Focusing on biodegradability, acidification and indigenous microorganisms, Environ. Technol. Innov. 36 (2024) 103870, https://doi.org/10.1016/j.eti.2024.103870.
[60] J.V. Mercado, M. Koyama, K. Nakasaki, Short-term changes in the anaerobic digestion microbiome and biochemical pathways with changes in organic load, Sci. Total Environ. 813 (2022) 152585, https://doi.org/10.1016/j.scitotenv.2021.152585.
[61] H. Wang, Q. Zhou, Bioelectrochemical anaerobic digestion mitigates microplastic pollution and promotes methane recovery of wastewater treatment in biofilm system, J. Hazard. Mater. 472 (2024) 134488, https://doi.org/10.1016/j.jhazmat.2024.134488.
[62] Y. Wang, L. Liang, J. Liu, D. Guo, Z. Zhu, H. Dong, Impact of anaerobic digestion on reactive nitrogen gas emissions from dairy slurry storage, J. Environ. Manag. 316 (2022) 115306, https://doi.org/10.1016/j.jenvman.2022.115306.
[63] S. Yang, W. Xue, P. Liu, X. Lu, X. Wu, L. Sun, et al., Revealing the methanogenic pathways for anaerobic digestion of key components in food waste: Performance, microbial community, and implications, Bioresour. Technol. 347 (2022) 126340, https://doi.org/10.1016/j.biortech.2021.126340.
[64] R.H. Kirkegaard, S.J. McIlroy, J.M. Kristensen, M. Nierychlo, S.M. Karst, M. S. Dueholm, et al., The impact of immigration on microbial community composition in full-scale anaerobic digesters, Sci. Rep. 7 (1) (2017) 9343, https://doi.org/10.1038/s41598-017-09303-0.
[65] L. Blasco, M. Kahala, S. Ervasti, E. Tampio, Dynamics of microbial community in response to co-feedstock composition in anaerobic digestion, Bioresour. Technol. 364 (2022) 128039, https://doi.org/10.1016/j.biortech.2022.128039.
[66] B.C. Diniz, P. Wilfert, D.Y. Sorokin, M.C.M. van Loosdrecht, Anaerobic digestion at high-pH and alkalinity for biomethane production: Insights into methane yield, biomethane purity, and process performance, Bioresour. Technol. 429 (2025) 132505, https://doi.org/10.1016/j.biortech.2025.132505.
[67] I. Vanwonterghem, P.N. Evans, D.H. Parks, P.D. Jensen, B.J. Woodcroft, P. Hugenholtz, et al., Methylotrophic methanogenesis discovered in the archaeal phylum Verstraetearchaeota, Nat. Microbiol. 1 (12) (2016) 16170, https://doi.org/10.1038/nmicrobiol.2016.170.
[68] USDA. Types of Anaerobic Digesters. U.S. Environmental Protection Agency 2025 June 24, 2025 ⟨https://www.epa.gov/anaerobic-digestion/types-anaerobic-digesters⟩, [cited 2026 2/13].
[69] L. Chen, W. Fang, J. Liang, M. Nabi, Y. Cai, Q. Wang, et al., Biochar application in anaerobic digestion: Performances, mechanisms, environmental assessment and circular economy, Resour. Conserv. Recycl. 188 (2023) 106720, https://doi.org/10.1016/j.resconrec.2022.106720.
[70] P. Zhang, D. Shen, J. Shao, X. He, J. Zeng, S.-L. Wu, et al., Green synthesis of Fe3O4@ceramsite from sludge improving anaerobic digestion performance of waste activated sludge, J. Environ. Manag. 359 (2024) 121085, https://doi.org/10.1016/j.jenvman.2024.121085.